Separating News from Entertainment: Attention and Political Accountability

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Abstract

How effective is public scrutiny of politicians when voters choose what media to consume? I model media technology as the cost of steering attention away from a platform's default bundle. Cheaper steering, which I call unbundling, lowers news consumption among voters who prefer entertainment and raises it among those who prefer news. When the bundle serves news and entertainment in equal proportions, these effects offset, so aggregate scrutiny falls only when voters on average prefer entertainment, and falls most when they share a moderate preference for it. Unbundling benefits every voter privately yet can reduce welfare, because each voter internalizes only a small share of the resulting fall in accountability.[1]

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1 Introduction

When news and entertainment are consumed together, exposure to political information is partly incidental. Back when television offered only a few channels, a viewer waiting for the evening film often had to sit through the news first. In the digital era, viewers can watch films on demand on streaming platforms. Therefore, the cost of adjusting media consumption to the individually optimal level has fallen, and exposure to less-preferred content has become elective. Does political accountability suffer when voters no longer have to sit through the news?

Media are one of the main channels through which voters hold politicians to account. Where newspapers circulate more widely, governments respond more to local needs (Besley and Burgess 2002), and where coverage of a legislator is thinner, voters know less about her and she works less for her constituency (Snyder Jr and Strömberg 2010). The digital transition has transformed this channel. The spread of broadband internet lowered turnout in local elections and changed local policy (Gavazza et al. 2019), and online outlets have come to substitute for traditional ones (Gentzkow 2007).

Choice is a very important feature of this transformation. Prior (2005) shows that as media options multiply, a person's relative preference for entertainment over news becomes a far stronger predictor of how much he knows about politics and whether he votes: those who like news consume more of it, those who prefer entertainment abandon it, and the gap between them widens. It is less clear what are the implications for political accountability. If some voters follow politics more closely while others stop, does scrutiny of incumbents rise or fall as a result, and do their incentives to exert effort change? I show that scrutiny can weaken once exposure to news is a matter of choice, and voters, on average, prefer entertainment. Accountability then depends less on the availability of political information than on the composition of the demand for it.

I study a two-period electoral accountability model with heterogeneous voters, a strategic incumbent, and free entry by media platforms. Each voter divides attention between news and entertainment, and differs in his intrinsic taste for news relative to entertainment. Every media firm supplies some default mixture of news and entertainment, and a voter who wants a diet different from that default exerts a *steering cost*, proportional to the informational distance between his preferred diet and the default. A bundled environment — a newspaper, a broadcast bulletin — has a high steering cost: editorial choices are made on the reader's behalf, and departing from them is relatively expensive. An unbundled environment — a streaming platform, a news website — has a low steering cost: the reader assembles his own mix almost freely.

Attention to the news produces scrutiny. Aggregate scrutiny disciplines the incumbent, who exerts costly first-period effort, and selects between types at the election. The resulting public good is non-rival: each voter enjoys it undivided but internalizes only a small fraction of his own contribution to it. On the supply side, news is produced whenever attention is larger than a critical mass determined by fixed newsroom costs, and in that case, every voter consumes his preferred amount of news.

The first result characterizes the *technology gap*: the change in a voter's news consumption when the environment moves from bundled to unbundled. There is no gap for voters with

no interest in news, for voters interested in nothing else, and for voters with tastes who consume the default mix under either technology. In other cases, unbundling reduces the news consumption of every entertainment-leaning voter and raises that of every news-leaning one, whatever the default. The technology gap is the largest not at the extremes but at intermediate tastes.

Suppose the default is evenly mixed, and that for every voter who prefers entertainment there is another who prefers news just as strongly. If cost of steering become cheaper, the aggregate scrutiny does not change, as the increase in news consumption among the news-preferring group compensates the decrease among entertainment-preferring group. Scrutiny then falls only when all voters, on average, prefer entertainment to news. An uneven default changes this result: if the default is news-heavy ($> 50\%$ of content devoted to news) then even a balanced electorate loses scrutiny, and with an entertainment-heavy one it gains. The size of the fall in news/entertainment depends also on the distribution of tastes. The largest possible loss occurs when every voter shares the same moderate preference for entertainment. If the default is evenly mixed, voters with balanced tastes do not change their consumption pattern, because they do not need to steer either to news or entertainment. Also, voters with extreme taste for entertainment consume little news either way.

On the incumbent's side, greater scrutiny always raises effort: more monitoring makes all incumbent motivated to exert effort and results in more first-period public good. However, at the same time, this response makes types harder to tell apart, because bad incumbents with for whom cost of effort is relatively low imitate good ones, and the share doing so grows linearly in scrutiny. I show that the informational advantage of a good incumbent is single-peaked, maximized at an intermediate level of attention — a "Goldilocks" region (Prato and Wolton 2016). Beyond that point, monitoring strengthens incentives and weakens selection at the same time.

When news is separated from entertainment, the private payoff increases for every voter: a lower steering price enables each to consume a media diet closer to his own taste, and only voters with exactly balanced tastes are left indifferent. While each voter has a limited impact on the public good, he enjoys it undivided — scrutiny is non-rival, yet his own contribution to it is diluted at a rate $1/N$, where N is the number of voters. He therefore receives the full private gain but bears only a $1/N$ share of the collective loss, so each voter responds to cheaper steering more than is socially optimal. Unbundling consequently faces no private opposition: no voter would choose to keep the bundle unless it happens to be his favorite content mix. Aggregate welfare can nonetheless fall when voters, on average, prefer entertainment, and within the family of taste distributions I study it falls at lower political stakes the more alike voters are.

2 Related literature

The paper formalizes the idea that much political learning is a by-product of consumption of other media contents (Downs, 1957). Prior (2007) documents that, as media choice expanded, the relative preference for entertainment became a strong predictor of political knowledge and participation, and Baum (2002) shows how soft news carries political information to otherwise inattentive audiences. Changes in the media environment have moved participa-

tion in the direction this view predicts: television lowered turnout by displacing news-rich media (Gentzkow, 2006), and so did broadband internet (Falck et al. 2014; Gavazza et al. 2019). In the model, what changes is not the supply of political information but the cost of avoiding it, which is one of the most salient features of the digital media.

This paper contributes to the theoretical literature on media and political accountability. Strömberg (2004) shows that media cover issues of interest to large audiences, so that policy favors groups that consume news; Besley and Prat (2006) study how media capture weakens accountability; and Prat and Strömberg (2013) survey the related literature. Closest in spirit, Hu et al. (2023) study personalized news aggregation for rationally inattentive voters. In these models, media firms decide about the supply. Here media firms follow demand and are not strategic players. The consumer makes the choice: news is available, but in an unbundled environment he can avoid it more easily. This also distinguishes the paper from models in which content depends on ideological or reputational incentives (Gentzkow and Shapiro 2006; Mullainathan and Shleifer 2005).

A large body of work studies how limited voter attention, often modeled through rational inattention, affects political outcomes: the implementation of reforms (Prato and Wolton 2018; Hu and Li 2019; Devdariani and Hirsch 2023; Blumenthal 2023; Blumenthal 2025), pandering incentives (Trombetta, 2020), polarization (Hu et al. 2023; Matějka and Tabellini 2021), and electoral outcomes (Martinelli 2006; Bruns and Himmler 2016); see Maćkowiak et al. (2023) and Balles et al. (2025) for reviews. The steering cost used here is the relative-entropy control cost of Mattsson and Weibull (2002), which also underlies the default-weighted logit of rational inattention (Matějka and McKay, 2015). In those models, acquiring information comes with a cost. Here information is free, and departing from the bundle of content a medium supplies by default is costly. The price of choosing is then an important primitive, not the price of becoming informed. This has three consequences which are different from the results of models of costly information. First, unbundling can make every voter better off privately while making the electorate worse off as a whole. Second, with an evenly mixed default, what matters for aggregate scrutiny is whether voters on average prefer entertainment or news, not how divided they are: an electorate split evenly between news and entertainment lovers is unaffected however strongly they disagree. Third, the voters with the largest changes of news consumption are not those with the strongest tastes but those with moderate ones.

The paper also relates to the literature on political agency. In Ferejohn (1986), elections discipline incumbents by threatening to remove poor performers. Fearon (1999) and Besley (2006) distinguish this sanctioning role from the selection of good types, a distinction employed in the analysis of Section 7. Ashworth (2012) and Duggan and Martinelli (2017) survey the related literature. Prato and Wolton (2016) and Ashworth and Bueno de Mesquita (2014) show that better informed or more attentive voters need not be better served. Here more scrutiny always strengthens discipline and eventually weakens selection, and yet the public good still rises.

Finally, the paper relates to the economics of bundling and of media markets. Bundling can raise or lower welfare through its effects on prices and on the products supplied (Adams and Yellen 1976; Bakos and Brynjolfsson 1999), and unbundling cable television has been studied as a policy in its own right (Crawford and Yurukoglu, 2012). In those analyses the effects of unbundling are private. Here unbundling also changes a public good, because the content

a voter stops consuming is the information that disciplines politicians. On the supply side, which products a market sustains depends on how many people want them when production involves fixed costs, a "preference externality" (George and Waldfogel 2003; Waldfogel 2007). The critical mass of attention that sustains a newsroom computed here is an instance of this externality.

3 Model setting

The game lasts two periods. There are N voters indexed by $J = 1, \ldots, N$. Voters differ in their taste for news, $\alpha_J \in (0, 1)$, drawn independently from a distribution F with density f; the taste for entertainment is $1 - \alpha_J$. The general results hold for any F. Where a parametric family is needed, I use the Beta distribution with mean μ and concentration ϑ, $\alpha_J \sim \text{Beta}\big(\vartheta\mu, \vartheta(1-\mu)\big)$, where larger ϑ means more alike voters. In this setting, I assume voters consume only these two types of content. I interpret news as any content that provides information about an incumbent, and entertainment as everything else. In the baseline specification, voters are the same in all dimensions but the taste parameter α. Additionally, the model includes media producers and a ruler. The latter can be of two types: good or bad, with exogenous probability of being good γ. If she is good, her cost of putting in effort is zero. Otherwise, her cost is drawn from a uniform distribution: $c \sim U[0, 1]$. A ruler decides how much effort to put into both periods, and the more effort she puts in, the more likely she is to be re-elected.

3.1 Timing

In the first period, a ruler (she) is randomly selected, and her type is her private information (good: $\theta = g$; or bad: $\theta = b$). Each consumer J (he) draws his taste parameter α_J and chooses the amount of entertainment $t^*_{J,e}$ and news $t^*_{J,n}$ to consume, with an "attention budget" constraint: $t^*_{J,n} + t^*_{J,e} = 1$. The average demands for entertainment and news are given by $t^*_e = \frac{\sum_{J=1}^{N} t^*_{J,e}}{N}$ and $t^*_n = \frac{\sum_{J=1}^{N} t^*_{J,n}}{N}$, respectively. Subsequently, M media producers decide to enter the market. Their number depends on the demand for media content, on exogenously determined advertising revenues, and on fixed costs for entertainment and news. Each media producer operates under the same profit function, and they cannot target specific consumers (i.e., their offer is homogeneous). Their choice is discrete: whether to fund a newsroom (i.e., produce news on top of entertainment). News is supplied exclusively by media firms; where a newsroom exists, news is freely available, so no consumer is rationed in news.

Then, if a ruler is of a bad type, she observes consumer and media firms choices and chooses the optimal amount of effort she puts in the first e_1 and second e_2 period, with $e_p \in [0, 1]$.[2] The public good in period p is proportional to the incumbent's effort: $e_p\tau_p$. It is non-rival and enters into consumers' utility function. Public-good provision depends on political tenure rather than calendar time. Only a re-elected incumbent can produce τ_2. A newly elected politician, even if entering in the second calendar period, is a first-term officeholder and can

[2] There is no difference if the ruler chooses the effort consecutively, as after the second period there are no re-election incentives. A good incumbent will choose maximum effort in the second period, and a bad incumbent will choose zero effort, regardless of the timing of the decision.

produce public good τ_1.

Subsequently, consumers consume media content. The more they pay attention to the news, the more motivated the incumbent is to exert effort. The probability of re-election is given by

$$\rho(\tilde{t}_n, e_1) = \tilde{t}_n\sqrt{e_1} + (1 - \tilde{t}_n)\rho_0 \tag{1}$$

where ρ_0 is the probability of re-election if a voter is not informed. Higher average news consumption improves voter monitoring and therefore increases the probability that poor-performing incumbents are replaced, while having a smaller effect on high-performing incumbents. Appendix A shows the microfoundation underlying this specification. Entertainment is also available from outside the modeled media sector, so no consumer is rationed in entertainment either.

Finally, elections are held during which consumers decide whether to re-elect the incumbent. If they do, the second period of the game begins, in which the politician puts effort e_2 into producing the public good. Note that consumers only obtain a noisy signal through consuming news about the incumbent's performance; the revelation of the realized amount of public good, $\tau_1 e_1 + \tau_2 e_2$, happens after the game is concluded.

3.2 Consumers/Voters

There are N voters indexed by $J = 1, \ldots, N$, each allocating a unit attention budget between news, $t_{J,n} \in [0, 1]$, and entertainment, $1 - t_{J,n}$. A media diet $t_{J,n}$ delivers content utility:

$$V(t_{J,n}; \alpha_J) = t_{J,n} \ln \alpha_J + (1 - t_{J,n}) \ln(1 - \alpha_J) = \ln(1 - \alpha_J) + x_J\, t_{J,n}, \tag{2}$$

where $x_J \equiv \ln \frac{\alpha_J}{1-\alpha_J}$ is the taste log-odds.

Media technology: a default bundle and a price of steering. A media environment is summarized by two objects. The first is its *default bundle* $\beta \in (0, 1)$: the share of news a viewer receives if he does not change the diet (e.g., the composition of the broadcast schedule, or of the feed he is served). The second is the *price of steering* $\kappa > 0$: changing the diet from the default requires seeking out content, resisting what is served, and switching, and κ represents the price of that effort. Formally, the cost of a diet $t_{J,n}$ is its deviation from the default:

$$C_{\kappa,\beta}(t) \;=\; \kappa\Big[t \ln \tfrac{t}{\beta} + (1 - t) \ln \tfrac{1-t}{1-\beta}\Big] \;\geq\; 0, \tag{3}$$

which vanishes only at $t = \beta$. Going with the flow (the default bundle) is free; departing from the default is costly in proportion to how far the diet departs from it. The steering cost (3) shares its functional form with the costs used in the rational-inattention literature, but is used here in a different context.

The public good. Apart from content value net steering costs, a consumer gets utility from the non-rival public good produced by an incumbent (which I interpret as *governance quality*). Incumbent's effort e_p in period p delivers a public good $e_p\tau_p$. Denote $g(\tilde{t}_n)$ for its expected total over the two periods, taken over the incumbent's type and cost draw. Aggregate scrutiny raises public good in two ways: it disciplines a bad incumbent into

exerting effort, and it raises the probability that a good one is retained. Two features of this public good influence the voter's problem.

First, every voter enjoys the public good delivered, which is undivided and does not depend on his own media diet. The level of public good is the same for an entertainment lover and a voter who follows every political story. Second, while the benefit is undivided, a voter's influence over it is not. His own attention impacts aggregate scrutiny only at a rate $1/N$, so the effect of his diet on $g(\tilde{t}_n)$ is of order $1/N$. In other words, benefits are shared in full by voters while influence is diluted. Monitoring of a politician is not their objective; it is only a by-product of preferences for media content. Formally, the expected public good over two periods is:

$$\begin{aligned} g(\tilde{t}_n) \;=\; \gamma\Big[\underbrace{\rho_g(\tilde{t}_n, e_1^*=1)\;\times(\tau_1+\tau_2)}_{\substack{\text{good, re-elected:}\\ \text{works both periods}}} \;+\; \underbrace{(1-\rho_g(\tilde{t}_n, e_1^*=1))\times\;(\tau_1+\gamma\tau_1)}_{\substack{\text{good, replaced: works in } t{=}1,\\ \text{challenger delivers } \gamma\tau_1}}\Big] \\ +\,(1-\gamma)\,\mathbb{E}_c\Big[\underbrace{\rho_b(\tilde{t}_n, e_1^*)\;\times\;\tau_1 e_1^*}_{\substack{\text{bad, re-elected:}\\ \text{shirks in } t{=}2}} \;+\; \underbrace{(1-\rho_b(\tilde{t}_n, e_1^*))\times\;(\tau_1 e_1^*+\gamma\tau_1)}_{\substack{\text{bad, replaced:}\\ \text{challenger delivers } \gamma\tau_1}}\Big], \end{aligned} \tag{4}$$

which simplifies to

$$g(\tilde{t}_n) = \gamma\Big[\tau_1+\rho_g(\tilde{t}_n, e_1^*=1)\times\tau_2+(1-\rho_g(\tilde{t}_n, e_1^*=1))\times\gamma\tau_1\Big]+(1-\gamma)\Big[\tau_1\,\mathbb{E}_c[e_1^*]+\big(1-\mathbb{E}_c[\rho_b]\big)\gamma\tau_1\Big]. \tag{5}$$

where γ is the exogenous probability of an incumbent being of a good type ($\theta = g$), $\rho_\theta(\tilde{t}_n, e_1^*)$ is the endogenous probability of re-election of an incumbent of type θ, and c is the cost of exerting an effort by a bad incumbent. Appendix B studies the extension in which voters differ in their own valuation of the public good $g(\tilde{t}_n)$.

The voter's payoff in full. Voter J chooses his news share $t_{J,n} \in (0,1)$ to maximize

$$W_J \;=\; \underbrace{t_{J,n}\ln\alpha_J + (1-t_{J,n})\ln(1-\alpha_J)}_{\text{content value}} \;-\; \underbrace{\kappa\Big[t_{J,n}\ln\tfrac{t_{J,n}}{\beta} + (1-t_{J,n})\ln\tfrac{1-t_{J,n}}{1-\beta}\Big]}_{\text{steering cost } C_{\kappa,\beta}(t_{J,n})} \;+\; \underbrace{g(\tilde{t}_n)}_{\text{public good}}, \tag{6}$$

s.t.

$$t_{J,n} \in [0,1], \quad \kappa > 0, \quad \beta \in (0,1). \tag{7}$$

Content value and the public good enter (6) additively, so they must be measured in the same units. I adopt the normalization that τ_1 and τ_2 are expressed in units of content utility: τ_p is the value, to a voter, of the public good an incumbent can deliver in period p, measured as the change in content value that would compensate him for it. The scale of τ does not affect the incumbent, whose choice depends on her rent, her cost of effort, and aggregate scrutiny, but not on τ. Nor, in the large-electorate limit, does it affect voters' diets or aggregate scrutiny, because each voter's influence on the public good does not play a role as the civic term becomes zero. In smaller electorates, at finite N, the effect is of order $1/N$. Hence, the normalization matters only for welfare comparisons, and Section 6 makes explicit which welfare results depend on it.

3.3 Media firms

Media firms sell attention to advertisers, earning A_n per unit of attention devoted to news and $A_e < A_n$ per unit devoted to entertainment.[3] Platforms produce both content types. The bundling of content is captured on the demand side by the steering cost (3). A platform's content decision is therefore a discrete one: whether to fund a newsroom, at fixed cost FC_n, in addition to the fixed cost FC_e of producing entertainment. Entertainment can be licensed or replicated cheaply, whereas reporting requires a newsroom whose cost does not scale with the audience. Platforms cannot target consumers, so each of the M platforms in the market serves an equal share N/M of voters. Entry is free. Section 4.3 derives the number of platforms and conditions for funding a newsroom.

3.4 Incumbent

I assume that a ruler has risk-neutral preferences and knows how much news voters consume. In each period in office, she receives a remuneration of r. If an elected politician is good, she faces no costs of exerting effort, $c = 0$. If she is of a bad type, her cost is drawn from a uniform distribution $c \sim U(0,1)$ (before solving the optimization problem). Each type chooses an amount of effort to put in both periods:

$$\begin{aligned} e_1^*, e_2^* = \arg\max_{e_1,e_2} \{(1 - c(\theta)e_1)r + (1 - c(\theta)e_2)r\rho(\tilde{t}_n, e_1)\} \\ \text{s.t.} \\ e_1, e_2 \in [0,1]; \quad r > 0 \\ \rho(\tilde{t}_n, e_1) = \tilde{t}_n\sqrt{e_1} + (1 - \tilde{t}_n)\rho_0 \\ c(\theta = g) = 0; \quad c(\theta = b) \sim U(0,1) \end{aligned} \tag{8}$$

The game can equivalently be formulated as sequential. For the bad incumbent, since second-period effort does not affect re-election prospects, it yields no strategic benefit, implying $e_2^* = 0$. The bad incumbent, therefore, effectively chooses only first-period effort.

3.5 Elections

The incumbent is re-elected by majority rule. Voters observe neither her type nor her effort. Each voter instead learns, with probability equal to his news share, his own verdict on her first term: *pass* or *fail*. It is drawn independently across voters. A good incumbent always passes; a bad one who exerted effort e_1 passes with probability $p(e_1)$, where p is increasing and concave with $p(0) = 0$ and $p(1) = 1$, so a fail reveals a bad type. After learning a verdict, a voter updates his belief by Bayes' rule. An uninformed voter keeps the prior γ. Each votes for the incumbent if the expected second-period public good is higher under her than under a challenger drawn from the same prior, adjusted by an idiosyncratic assessment of the candidates, such as party loyalty or personal appeal, and by a common popularity shock.

[3] The premium reflects exclusivity: local reporting is produced by the modeled firms alone, whereas entertainment is a commodity available from many sources.

3.6 Equilibrium concept

I focus on pure strategies, and the equilibrium is a Perfect Bayesian Equilibrium. On an equilibrium path, the ruler chooses the level of effort, taking the attention of consumers to news as given, and consumers divide their attention between news and entertainment, correctly foreseeing the effort of an incumbent, given the prior γ and the distribution of c. Because payoffs depend only on realized consumption, I analyze equilibrium using a reduced game in realized choices. The appendix includes the definition of equilibrium along with proofs of its existence (Appendix C).

4 Solution

Equilibrium characterization. For every $\kappa > 0$, $\beta \in (0,1)$ and for every taste distribution $\alpha \sim F$, the game of Section 3 has a unique perfect Bayesian equilibrium, with the following properties.

(i) **Voters.** Each voter's demand for news $\tilde{t}_{J,n}$ is interior and satisfies

$$t^*_{J,n}(\alpha_J;\kappa,\beta,\tilde{t}_n) = \frac{\alpha_J^{1/\kappa}\,\beta\,e^{\zeta}}{\alpha_J^{1/\kappa}\,\beta\,e^{\zeta} + (1-\alpha_J)^{1/\kappa}(1-\beta)} \in (0,1), \qquad \zeta \equiv \frac{g'(\tilde{t}_n)}{N\kappa}, \tag{9}$$

increasing in his taste α_J, in the default β, and in the civic term ζ, which measures the marginal effect of his own attention on the public good. Since ζ is of order $1/N$, this term is small in a large electorate, and (9) reduces to a function of (α_J,κ,β) alone.

(ii) **Media firms.** Free entry determines the number of platforms M and the break-even level of demand for news $\bar{d}$. Where news is produced, no voter is rationed, so realized average scrutiny solves the fixed point

$$\tilde{t}_n = \begin{cases} \mathbb{E}_F\left[t^*_n(\alpha;\kappa,\beta,\tilde{t}_n)\right] & \text{if } \mathbb{E}_F[t^*_n] \geq \bar{d}, \\ 0 & \text{otherwise.} \end{cases} \tag{10}$$

(iii) **Incumbent.** A good type exerts full effort in both terms; a bad type with cost c exerts $e^*_1(c;\tilde{t}_n)$ of (13) in the first term and none in the second.

(iv) **Election.** The incumbent is re-elected with probability (1); hence (12) for a good type and (15) in expectation for a bad one.

(v) **Consistency.** Voters' expectations are correct: the marginal public good $g'(\tilde{t}_n)$ entering (22) is evaluated at the incumbent's equilibrium effort from (iii).

I solve the model using backward induction. Because voters' diets depend on aggregate scrutiny only through $g'(\tilde{t}_n)$, and the incumbent's effort depends on $\tilde{t}_n$ alone, I first solve the election, then the incumbent's problem, taking realized average news as given, then the media market and the voters' problem, and close the model with (10).

4.1 The election

In the second term there are no re-election incentives, so a bad officeholder exerts no effort and a good one exerts full effort, whether she is the incumbent or newly elected. Re-electing a good incumbent therefore delivers τ_2, re-electing a bad one delivers nothing, and a challenger delivers τ_1 with probability γ. Given the effort bad incumbents exert in equilibrium, Bayes' rule gives informed voters the beliefs

$$\pi_{\text{pass}} = \frac{\gamma}{\gamma + (1-\gamma)\,\bar{p}_b} > \gamma, \qquad \pi_{\text{fail}} = 0, \qquad \bar{p}_b \equiv \mathbb{E}_c\big[p(e_1^*(c))\big], \tag{11}$$

while uninformed voters keep $\pi_U = \gamma$. A voter with belief π gains $B(\pi) = \pi\tau_2 - \gamma\tau_1$ from re-electing the incumbent, so

$$B_{\text{fail}} = -\gamma\tau_1 \;<\; B_U = \gamma(\tau_2 - \tau_1) \;<\; B_{\text{pass}} = \pi_{\text{pass}}\tau_2 - \gamma\tau_1.$$

Appendix A shows that, with uniform idiosyncratic assessments and a uniform popularity shock, majority rule yields

$$\rho(\tilde{t}_n, e_1) = \tilde{t}_n\,\Gamma\big(B_I(e_1)\big) + (1-\tilde{t}_n)\,\Gamma(B_U), \qquad B_I(e_1) = p(e_1)B_{\text{pass}} + \big(1-p(e_1)\big)B_{\text{fail}},$$

where B_I is the average benefit among informed voters and $\Gamma(X) = \Pr(\xi > -X)$ is the probability that the incumbent is re-elected when voters' average benefit from re-electing her is X. Γ is increasing and affine because the shock is uniform, so the probability of re-election at an average benefit equals the average of the corresponding probabilities. This has the form of (1): $q(e_1) = \Gamma\big(B_I(e_1)\big)$ is increasing and concave in effort, and $\rho_0 = \Gamma(B_U) > \Gamma(B_{\text{fail}}) = q(0)$, since uninformed voters think better of the incumbent than informed voters who saw a fail. The specification $q(e_1) = \sqrt{e_1}$ is a normalization of this structure.[4]

The incumbent takes (1) as given. Her effort matters only through informed voters, so its electoral return, $\partial\rho/\partial e_1 = \tilde{t}_n\, q'(e_1)$, is proportional to aggregate news consumption.

4.2 The incumbent's turn

The incumbent chooses first-period effort, taking aggregate scrutiny $\tilde{t}_n$ and the probability of re-election (1) as given. A good incumbent faces no cost of effort and exerts full effort, $e_1 = 1$, so she is re-elected with probability

$$\rho_g(\tilde{t}_n) = \tilde{t}_n + (1-\tilde{t}_n)\,\rho_0. \tag{12}$$

A bad incumbent with cost c exerts no effort in the second period (Section 4.1), so she chooses $e_1 \in [0,1]$ to maximize $r(1 - c\,e_1) + r\,\rho(\tilde{t}_n, e_1)$. The objective is concave in e_1, and its first-order condition, $\tilde{t}_n/(2\sqrt{e_1}) = c$, gives

$$e_1^*(c;\tilde{t}_n) = \min\left\{1, \left(\frac{\tilde{t}_n}{2c}\right)^2\right\}. \tag{13}$$

Effort is strictly positive for every c whenever $\tilde{t}_n > 0$, and is full exactly when $c \le \tilde{t}_n/2$; since $c \sim U(0,1)$, the share of bad incumbents exerting full effort is $\tilde{t}_n/2$. Integrating (13) over c,

$$\mathbb{E}_c[e_1^*] = \int_0^{\tilde{t}_n/2} dc + \int_{\tilde{t}_n/2}^1 \left(\frac{\tilde{t}_n}{2c}\right)^2 dc = \tilde{t}_n - \frac{\tilde{t}_n^2}{4}, \tag{14}$$

[4] Appendix A shows that it implies $\rho_0 = \gamma + (1-\gamma)\bar{p}_b \ge \gamma$, that q and ρ_0 cannot both be held fixed as scrutiny varies, so the analysis treats them as a reduced form.

which is strictly increasing on $[0,1]$, with slope $1-\tilde{t}_n/2$: scrutiny always disciplines. Similarly, since $\mathbb{E}_c[\sqrt{e_1^*}] = \int_0^{\tilde{t}_n/2} dc + \int_{\tilde{t}_n/2}^1 \frac{\tilde{t}_n}{2c}\, dc = \frac{\tilde{t}_n}{2}\left(1 - \ln\frac{\tilde{t}_n}{2}\right)$, a bad incumbent is re-elected with expected probability

$$\mathbb{E}_c[\rho_b] = \frac{\tilde{t}_n^2}{2}\left(1 - \ln\frac{\tilde{t}_n}{2}\right) + (1-\tilde{t}_n)\,\rho_0. \tag{15}$$

Figure 1 plots (13) and (14). The gap between $\mathbb{E}_c[e_1^*]$ and the 45-degree line illustrates diminishing returns to scrutiny: each additional unit of attention to news induces less effort from a bad incumbent, because a share $\tilde{t}_n/2$ of bad incumbents already exert full effort and cannot respond further.

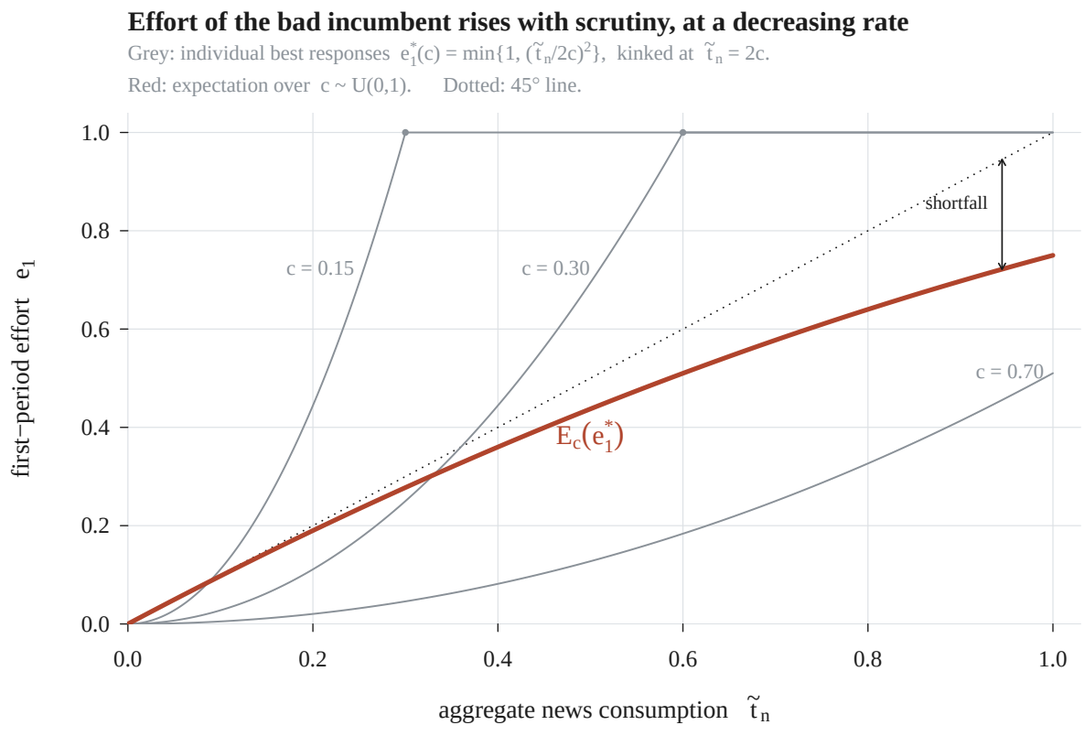


Figure 1: Best response of the bad incumbent. Grey curves are individual best responses (13) for selected costs c; the red curve is the expectation (14) over $c \sim U(0,1)$; the dotted line is the 45-degree line.

4.3 Media firms' turn

Platforms take demand for news as given. In the large-electorate limit, demand $d \equiv \mathbb{E}_F\left[t_n^*(\alpha;\kappa,\beta)\right]$ depends only on the technology, the default, and the distribution of tastes. Each of the M platforms serves an equal share N/M of voters. A platform without a newsroom earns A_e per unit of attention to entertainment; one with a newsroom earns additionally A_n on the share d devoted to news. Profits are

$$\pi_1(M) = \frac{N}{M}\left[A_e + (A_n - A_e)d\right] - FC_e - FC_n, \qquad \pi_0(M) = \frac{N}{M}\,A_e - FC_e. \tag{16}$$

Given the number of platforms, funding a newsroom pays if and only if $\pi_1(M) \geq \pi_0(M)$, that is

$$d \;\geq\; d(M) \equiv \frac{M\,FC_n}{(A_n - A_e)\,N}. \tag{17}$$

A newsroom does not create attention — voters have a unit budget either way — but converts the share d of it from the entertainment rate to the news rate. It is feasible when

the premium earned on a platform's audience covers the cost of reporting. The threshold $d(M)$ depends on the number of platforms, which is itself determined by entry.

Lemma 1 (Critical mass under free entry)**.** *There is a free-entry equilibrium in which news is produced if and only if $d \geq \bar{d}$, and one in which it is not if and only if $d \leq \bar{d}$, where*

$$\bar{d} \;=\; \frac{FC_n}{FC_e} \times \frac{A_e}{A_n - A_e}. \tag{18}$$

With news, $M = N[A_e + (A_n - A_e)d]/(FC_e + FC_n)$; without, $M = NA_e/FC_e$ (assuming M is continuous). The threshold $\bar{d}$ depends neither on the size of the market, nor on the number of platforms, nor on the media technology.

Proof. News equilibrium. Zero profit, $\pi_1(M) = 0$, gives $N/M = (FC_e + FC_n)/[A_e + (A_n - A_e)d]$. No platform gains by closing its newsroom iff (17) holds at this audience size, which rearranges to $FC_e(A_n - A_e)d \geq FC_nA_e$, that is $d \geq \bar{d}$. The same inequality ensures that an entrant without a newsroom, earning $(N/M)A_e - FC_e$, would make no profit.

No-news equilibrium. Zero profit, $\pi_0(M) = 0$, gives $N/M = FC_e/A_e$. No platform gains by funding a newsroom iff $(N/M)(A_n - A_e)d \leq FC_n$, that is $d \leq \bar{d}$; the same inequality ensures that an entrant with a newsroom would make no profit. □

Market size does not play a role because entry adjusts the number of platforms rather than their size. At $d = \bar{d}$ platforms are indifferent; I adopt the convention that they report.[5] Because news carries no marginal cost once a newsroom exists, a platform that reports makes news freely available, and no voter is rationed. Realized scrutiny is therefore

$$\tilde{t}_n = \begin{cases} d, & d \geq \bar{d}, \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

Platforms are demand-followers: they neither shape the composition of what voters consume nor ration it, and their only influence on accountability is by making a discrete decision of whether reporting exists at all. That decision makes the supply side a source of discontinuity. Denote d_B and d_U for demand for news under the bundled and unbundled technologies.

Proposition 2 (News-market collapse)**.** *Suppose news is viable under bundling, $d_B \geq \bar{d}$.*

(i) News production ceases after unbundling if and only if $\bar{d} \in (d_U, d_B]$.

(ii) This set is non-empty if and only if unbundling lowers demand for news, $d_U < d_B$, and it is then an interval of width $d_B - d_U$.

Proof. By Lemma 1, news is produced in an environment with demand d if and only if $d \geq \bar{d}$, and $\bar{d}$ is the same under both technologies. Production therefore occurs under bundling and ceases under unbundling if and only if $d_U < \bar{d} \leq d_B$, which is (i). Part (ii) follows directly. □

[5] With an integer number of platforms the equilibrium remains unique, and news is produced if and only if demand exceeds a threshold d_N that does not depend on the media technology, so Proposition 2 holds with d_N in place of $\bar{d}$. Integer effects let platforms earn rents, which lowers the threshold by at most $FC_n/[(A_n - A_e)N]$ whenever entertainment alone can sustain a platform, so $d_N \to \bar{d}$ as the market grows; see Appendix C, Step 4.

A fall in demand thus lowers scrutiny marginally if producing news remains viable, but removes it if demand falls below $\bar{d}$. Market size matters when entry cannot adjust. Because attention to news earns more than attention to entertainment, $A_n > A_e$, a platform that reports earns more per consumer than one that does not. In a market too small to support a platform without a newsroom, $NA_e < FC_e$, this extra revenue is what keeps local media alive: a single reporting platform is viable if and only if its revenue covers both fixed costs, $N\big[A_e + (A_n - A_e)d\big] \geq FC_e + FC_n$, that is,

$$d \;\geq\; \frac{(FC_e + FC_n)/N - A_e}{A_n - A_e}.$$

This threshold exceeds $\bar{d}$, rises as the market shrinks, and can be met at all only if $NA_n \geq FC_e + FC_n$. Below it, no platform is viable. In such places, such as small towns served by a single outlet, a fall in demand for news does not merely decrease local reporting: it might remove local media altogether.

4.4 Voters' turn

Voter J chooses his news share to maximize content value net of steering cost, internalizing the effect of his attention on the public good. The first-order condition equates the two marginal benefits of news with its marginal cost:

$$\underbrace{x_J}_{\substack{\text{content}\\ \text{value}}} + \underbrace{\frac{g'(\tilde{t}_n)}{N}}_{\substack{\text{own contribution}\\ \text{to the public good}}} = \underbrace{\kappa\Big[\ln\tfrac{t_{J,n}}{1-t_{J,n}} - \bar{b}\Big]}_{\substack{\text{marginal cost}\\ \text{of steering}}}. \tag{20}$$

A voter steers his attention toward the news until the taste gains from one more unit, together with the marginal scrutiny it produces, equals the effort of obtaining it. Rearranging (20) gives the model's main relation:

$$\boxed{\ln\frac{t^*_{J,n}}{1-t^*_{J,n}} \;=\; \underbrace{\frac{x_J}{\kappa}}_{\text{own taste}} + \underbrace{\bar{b}}_{\text{default}} + \underbrace{\zeta(\tilde{t}_n)}_{\text{civic term}}, \qquad \zeta \equiv \frac{g'(\tilde{t}_n)}{N\kappa}} \tag{21}$$

The log-odds of a voter's diet decompose additively into three terms: his own taste, scaled by the price of steering; where the environment pushes the attention; and the value of the scrutiny his attention produces. Only the first varies across voters — the default and the civic term are common — and the third is of order $1/N$, so that in a large electorate diets are mostly governed by tastes and the environment alone. Solving (21),

$$t^*_{J,n}(\alpha_J;\kappa,\beta,\tilde{t}_n) = \frac{\alpha_J^{\,1/\kappa}\,\beta\,e^{\zeta}}{\alpha_J^{\,1/\kappa}\,\beta\,e^{\zeta} + (1-\alpha_J)^{\,1/\kappa}(1-\beta)}. \tag{22}$$

The civic motive is observationally equivalent to facing a slightly more news-oriented default β: internalizing the public good and being served more news produce the same diets. Differentiating (21) with respect to the taste log-odds,

$$\frac{\partial t^*_{J,n}}{\partial x_J} = \frac{t^*_{J,n}(1-t^*_{J,n})}{\kappa}, \tag{23}$$

which measures how strongly a difference in taste translates into a difference in what voters actually consume. First, pass-through in (23) is inversely proportional to the price of steering. If steering is expensive, voters with very different tastes end up with similar media diets, because nobody finds it worth the effort to depart from what is served in default. If

steering is cheap, the same voters end up with widely differing diets. Therefore, the price of steering influences the rate at which heterogeneity in preferences converts into heterogeneity in consumption. It is this conversion rate, not preferences themselves, that the digital transition changed, defined here also as unbundling of media content. Second, voters differ in pass-through; the factor $t^*_{J,n}(1-t^*_{J,n})$ is largest for voters whose diets are close to balanced (who consume news and entertainment in similar proportions) and disappears in extreme cases, so the effect of unbundling is the weakest for the voters whose diets are already close to one content type.

Lemma 3 (Dual CES representation, inspired by Anderson et al. (1992))**.** *Fix $\kappa > 0$, $\beta \in (0,1)$ and ζ. Demand (22) coincides with that of a voter maximizing the CES objective $\big((1-a)(1-t)^q + a\,t^q\big)^{1/q}$ over $t \in (0,1)$, with substitution parameter $q = 1-\kappa$ and share parameter a satisfying*

$$\frac{a}{1-a} = \frac{\alpha_J}{1-\alpha_J} \times \left(\frac{\beta\, e^{\zeta}}{1-\beta}\right)^{\kappa}. \tag{24}$$

Proof. Let $q < 1$. The CES aggregate is strictly quasi-concave in t, and the optimum is interior: for $q \in (0,1)$ the marginal value of each good is unbounded at a zero share, and for $q < 0$ the aggregate vanishes at both corners. The optimum is therefore characterized by the first-order condition

$$a\,t^{q-1} = (1-a)(1-t)^{q-1}, \qquad \text{that is} \qquad \frac{t}{1-t} = \left(\frac{a}{1-a}\right)^{1/(1-q)}.$$

Setting $q = 1-\kappa$ and substituting (24),

$$\ln\frac{t}{1-t} = \frac{1}{\kappa}\ln\frac{a}{1-a} = \frac{x_J}{\kappa} + \bar{b} + \zeta,$$

which is (21). Both problems have a unique solution, so they deliver the same demand.[6] □

The lemma locates the model relative to the specification in which media technology is an elasticity of substitution in preferences: for instance, the bundled benchmark $\kappa = 2$ corresponds to CES complements ($q = -1$), the unbundled benchmark $\kappa = \frac{1}{2}$ to CES substitutes ($q = \frac{1}{2}$), and the two generate identical behavior. It also identifies the difficulty with that specification. Equation (24) holds the environment fixed; except in the symmetric case $\beta e^{\zeta} = \frac{1}{2}$, the share parameter representing a given voter depends on the technology. The same CES preference, therefore, cannot describe him in both environments, so a change in technology would have to be recorded as a change in his taste for news. The present formulation avoids this: tastes are the weights α_J, fixed across regimes, which are described by parameters (κ, β).

5 Effects of unbundling of news and entertainment

5.1 Technology gap

This section discusses the *technology gap*, which I define as the change in attention of a voter to news when the environment shifts from bundled (interpreted as traditional media) to

[6] At $\kappa = 1$, so $q = 0$, the CES aggregate is replaced by its Cobb–Douglas limit $a\ln t + (1-a)\ln(1-t)$, whose first-order condition $a/t = (1-a)/(1-t)$ is the case $q = 0$ of the one above; the argument is unchanged.

unbundled (digital media). This subsection analyses individual technology gaps, subsection 5.2 describes aggregation across voters and its consequences for overall scrutiny. Lemma 4 evaluates diets in the large-electorate limit, where the civic term vanishes. A voter's diet reflects two motives: his taste for content, and the civic value of the scrutiny his attention produces. The second is small in a large electorate, because each voter's attention contributes only $1/N$ of aggregate scrutiny. Precisely, g' is continuous on $[0,1]$ and hence bounded, $|g'| \leq \bar{g}$, so $\zeta = g'(\tilde{t}_n)/(N\kappa)$ is at most $\bar{g}/(N\kappa)$, whatever the level of scrutiny. Since the slope of Λ is at most $\frac{1}{4}$, the civic term raises a voter's news share by at most $\bar{g}/(4N\kappa)$ in each environment, so his technology gap differs from $h(\alpha)$ by at most $\frac{\bar{g}}{4N}\left(\frac{1}{\kappa_U} + \frac{1}{\kappa_B}\right)$. The lemma therefore isolates the content motive, which is what the technology changes. Remark 5 describes what the civic term does at finite N.

Lemma 4 (Shape of the technology gap)**.** *Let $\kappa_B > \kappa_U > 0$ and $\beta \in (0,1)$. Diets are evaluated in the large-electorate limit, where the civic term $\zeta = 0$, so that $t_n^*(\alpha;\kappa,\beta)$ depends on the environment only through (κ,β). Define the technology gap*

$$h(\alpha) \;\equiv\; t_n^*(\alpha;\kappa_U,\beta) - t_n^*(\alpha;\kappa_B,\beta).$$

Then:

(i) $h(0) = h(\frac{1}{2}) = h(1) = 0$, and these are the only zeros. The gap becomes zero in the extreme cases — no interest ($\alpha = 0$) or exclusive interest ($\alpha = 1$) — and for perfectly balanced tastes ($\alpha = \frac{1}{2}$).

(ii) $h < 0$ on $(0, \frac{1}{2})$ and $h > 0$ on $(\frac{1}{2}, 1)$: unbundling reduces the news consumption of voters who, on average, prefer entertainment and raises that of voters who, on average, prefer news, whatever the default bundle β.

(iii) Write $h(\alpha;\beta)$ to display the dependence on the default. For every $\alpha, \beta \in (0,1)$,

$$h(1-\alpha;\, 1-\beta) \;=\; -\,h(\alpha;\, \beta).$$

Interchanging the roles of news and entertainment — in the voter's taste and in the default bundle — reverses the technology gap.

(iv) Suppose the bundled technology compresses tastes while the unbundled one amplifies them, $\kappa_B \geq 1 \geq \kappa_U$ with at least one inequality strict. If the default mix is symmetric, $\beta = \frac{1}{2}$, then h is strictly convex on $(0, \frac{1}{2})$ with a unique interior minimizer $\alpha^\dagger$, and strictly concave on $(\frac{1}{2}, 1)$ with a unique maximizer at $1 - \alpha^\dagger$.

Proof in Appendix D.

Remark 5 (The civic term at finite N). Lemma 4 sets the civic term to zero. With N finite, $\zeta_\kappa = g'(\tilde{t}_n)/(N\kappa) > 0$ differs across environments, and the technology gap becomes

$$h_N(\alpha) = \Lambda\Big(\tfrac{x}{\kappa_U} + \bar{b} + \zeta_U\Big) - \Lambda\Big(\tfrac{x}{\kappa_B} + \bar{b} + \zeta_B\Big), \qquad x = \ln\tfrac{\alpha}{1-\alpha}, \qquad \Lambda(z) = 1/(1+e^{-z}). \tag{25}$$

Suppose $\zeta_U > \zeta_B$, which holds whenever $g'(\tilde{t}_n^U)/g'(\tilde{t}_n^B) > \kappa_U/\kappa_B$, and in particular whenever unbundling does not raise scrutiny, since g is concave. Parts (i) and (ii) of the lemma then still hold with the midpoint replaced by a nearby taste. Because Λ is strictly increasing, $h_N(\alpha) = 0$ at an interior taste if and only if the two arguments are equal, which happens at the single log-odds

$$x_N = -\,(\zeta_U - \zeta_B)\,\frac{\kappa_B\kappa_U}{\kappa_B - \kappa_U} < 0, \tag{26}$$

whatever the default. The interior zero thus moves from $\frac{1}{2}$ to $\alpha_N = \Lambda(x_N) < \frac{1}{2}$, with $|\alpha_N - \frac{1}{2}| \leq |x_N|/4$; the zeros at $\alpha = 0$ and $\alpha = 1$ remain; and $h_N < 0$ on $(0, \alpha_N)$ and $h_N > 0$ on $(\alpha_N, 1)$. Part (iii) does not hold anymore: the civic term always increases demand for news, so interchanging news and entertainment no longer reverses the gap exactly.

Taste parameter α determines the magnitude of change in consumed news after transitioning from traditional (bundled) to digital (unbundled) media. On the left side in Figure 2, taking the symmetric default $\beta = \frac{1}{2}$, the bundled response curve lies above the 45-degree line on $(0, \frac{1}{2})$ and is concave there, while the unbundled curve lies below it and is convex. On the right, their difference $h(\alpha) = t_U^* - t_B^*$ is therefore convex on $(0, \frac{1}{2})$, negative below the midpoint and positive above it, with a unique minimizer at $\alpha^\dagger \approx 0.20$ where a voter changes about 0.275 of his attention budget from news to entertainment.

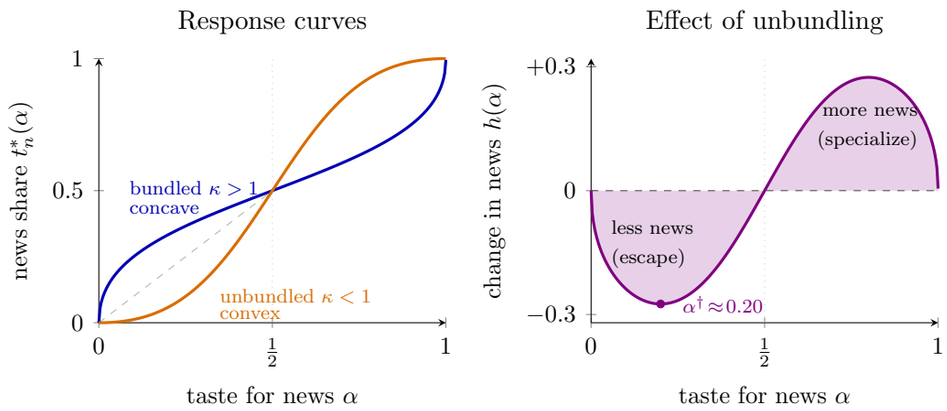


Figure 2: **Effects of unbundling, symmetric default** ($\beta = \frac{1}{2}$)

In the next subsection, I describe the aggregate effects for any $\kappa_B > \kappa_U$, that is, for any case when unbundling made the cost of steering cheaper for voters.

5.2 Aggregation: the role of the distribution of preferences

Assume in this section that the cost of steering is higher in a bundled than in an unbundled environment, that is, $\kappa_B > \kappa_U$. The technology gap h describes how one voter responds to unbundling, but what matters for accountability is the response of the electorate. Define the aggregate effect:

$$\Delta(F) \equiv \int_0^1 h(\alpha)\, f(\alpha)\, d\alpha = \mathbb{E}_F[t_n^*(\alpha; \kappa_U, \beta)] - \mathbb{E}_F[t_n^*(\alpha; \kappa_B, \beta)], \tag{27}$$

as the change in average news consumption when the environment moves from bundled to unbundled, holding tastes fixed. Since N tastes denoted by α are i.i.d draws, (27) is the expected change for every N. Because h is negative below $\frac{1}{2}$ and positive above, $\Delta(F)$ combines a loss of scrutiny among voters who prefer entertainment with a gain among those who prefer news, and which of the two dominates is not obvious in advance.

The analysis becomes more intuitive if we take mirror-image pairs from the electorate. Consider two voters with tastes α and $1-\alpha$: one prefers entertainment by exactly as much as the other prefers news. If the default bundle is symmetric ($\beta = \frac{1}{2}$), Lemma 4(iii) says unbundling treats them as exact opposites, $h(1-\alpha) = -h(\alpha)$: the first loses precisely as much news as the second gains, so the pair contributes nothing to the aggregate. If there are the same

number of voters on the two sides of every pair of tastes, all such contributions would cancel and $\Delta(F)$ would be zero regardless of the size of the individual responses. Substituting α with $1-\alpha$ on $(\frac{1}{2},1)$ in (27) makes the cancellation explicit:

$$\Delta(F) = \int_0^{1/2} h(\alpha)\left[f(\alpha) - f(1-\alpha)\right] d\alpha. \tag{28}$$

Expressions (27) and (28) are the same number, but the second highlights the impact of the imbalance in population weights, $f(\alpha)-f(1-\alpha)$, on the change in scrutiny. With an evenly mixed default, the aggregate therefore depends on the taste distribution only through its asymmetry around the midpoint $\frac{1}{2}$. The next proposition specifies what this implies, and what changes when the default is uneven.

Proposition 6 (Distributional effects of unbundling)**.** *Let $\kappa_B > \kappa_U > 0$ and $\beta \in (0,1)$. Parts (i)–(ii) hold for every default mix β; part (iii) is stated for the Beta family; part (iv) requires $\beta = \frac{1}{2}$ and $\kappa_B \geq 1 \geq \kappa_U$.*

(i) ***One-sided electorates.*** *If no voter prefers news over entertainment — F supported on $(0, \frac{1}{2}]$ and not degenerate at $\frac{1}{2}$ — then $\Delta(F) < 0$. Symmetrically, if no voter leans towards entertainment, $\Delta(F) > 0$. The default mix β plays no role.*

(ii) ***Balanced electorates and the default.*** *If the CDF of voters' tastes F is symmetric around $\frac{1}{2}$ and F is not the point mass at $\frac{1}{2}$, then*

$$\operatorname{sign}\Delta(F) = \operatorname{sign}\left(\tfrac{1}{2} - \beta\right),$$

whatever the dispersion of F. Neutrality (zero technology gap), therefore, requires two symmetries, of tastes and of the default: $\Delta(F) = 0$ for every symmetric F if and only if $\beta = \frac{1}{2}$. An entertainment-leaning default ($\beta < \frac{1}{2}$) leaves a balanced electorate consuming more news after unbundling, because bundling was holding it below its preferred mix. Moreover $\Delta(\check{F}; 1-\beta) = -\Delta(F;\beta)$, so it suffices to study $\beta \leq \frac{1}{2}$.

(iii) ***Threshold.*** *Let $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1-\mu))$. Then $\Delta \to 0$ as $\mu \to 0$ and as $\mu \to 1$, and as a function of μ it changes sign at most once, necessarily from negative to positive. There is therefore at most one threshold $\mu^*(\beta,\vartheta)$, and where it exists, $\Delta < 0$ for $\mu < \mu^*$ and $\Delta > 0$ for $\mu > \mu^*$. When $\beta = \frac{1}{2}$ a threshold exists for every ϑ, and $\mu^* = \frac{1}{2}$. More generally, for every ϑ there is $\varepsilon(\vartheta) > 0$ such that a threshold exists whenever $|\beta - \frac{1}{2}| < \varepsilon(\vartheta)$.*

(iv) ***Homogeneity makes the loss larger.*** *Let $\beta = \frac{1}{2}$ and $\kappa_B \geq 1 \geq \kappa_U$, fix $\mu < \frac{1}{2}$, and suppose no voter leans towards news, so tastes lie in $(0, \frac{1}{2}]$. Then*

$$|\Delta| \;\leq\; \big|h(\mu)\big|,$$

with equality only if every voter has taste exactly μ, and any mean-preserving spread that keeps every taste in $(0, \frac{1}{2}]$ strictly reduces $|\Delta|$.

Proof in Appendix E.

For Beta electorates with an evenly mixed default, $\beta = \frac{1}{2}$, the threshold is the midpoint, $\mu^* = \frac{1}{2}$. An uneven default changes it, because it makes the two effects of unbundling unequal. A news-heavy default gives voters preferring entertainment a diet close to balanced,

where diets respond most to the technology, and leaves voters preferring news little room to add news; the loss then outweighs the gain, and scrutiny falls even in a balanced electorate. By Proposition 6(ii)–(iii), the threshold then lies above the midpoint: scrutiny can fall even when voters on average prefer news. An entertainment-heavy default reverses this. A balanced electorate gains scrutiny, and it falls only when average taste lies sufficiently far below the midpoint.

Figure 3(a) illustrates the argument behind Proposition 6(iv). The red curve is $h(\alpha)$, the gap faced by a single voter of taste α. The homogeneous electorate, in which every voter has taste 0.40, lies on the curve; the electorate placing one third of voters at 0.20 and two thirds at 0.50 has the same mean and lies on the chord. The vertical distance, 0.050, is the reduction in the fall of aggregate scrutiny. Among electorates with no news-leaning voters, heterogeneity is thus a form of insurance against a large drop in scrutiny. The same argument applies to an electorate that, on average, prefers entertainment, such as a Beta electorate with $\mu < \frac{1}{2}$, once mirror-image voters have canceled: the voters who remain lie below $\frac{1}{2}$.

Figure 3(b) shows the corresponding comparison within the Beta family, holding the mean at $\mu = 0.40$ at the benchmark technologies with an evenly mixed default ($\beta = 0.5$). The fall in scrutiny deepens with the concentration of tastes ϑ, approaching the homogeneous benchmark $|h(0.40)|$ from below. At low concentration the fall is small for a different reason than in panel (a): the Beta distribution then places most voters near 0 and 1, where diets barely respond to the technology. This comparison is numerical; it is not implied by Proposition 6(iv).

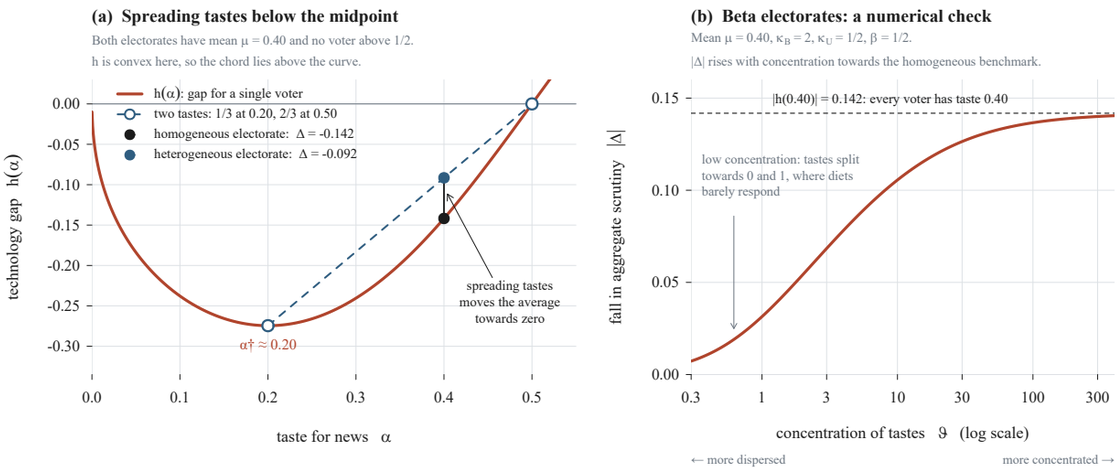


Figure 3: Distributional effects of unbundling. Panel (a): the technology gap $h(\alpha)$, the homogeneous electorate (every voter at 0.40, on the curve) and an electorate with the same mean placing one third of voters at 0.20 and two thirds at 0.50 (on the chord). Panel (b): the fall in aggregate scrutiny for Beta electorates with mean $\mu = 0.40$, as a function of the concentration of tastes ϑ. Parameters: $\kappa_B = 2$, $\kappa_U = \frac{1}{2}$, $\beta = \frac{1}{2}$.

6 Voters' welfare

Proposition 6 gives the conditions under which unbundling lowers aggregate scrutiny, defined as the average demand for news. It does not follow that welfare falls: voters value content as well as accountability, and the change in technology affects both. This section separates the two channels and shows that they move against each other exactly when unbundling lowers

scrutiny, which with an evenly mixed default happens when the electorate leans towards entertainment.

Recall from Section 4 that a voter with taste α facing steering price κ gets indirect utility

$$v(\alpha;\kappa,\beta) = \kappa \ln\Big[\beta\,\alpha^{1/\kappa} + (1-\beta)(1-\alpha)^{1/\kappa}\Big], \tag{29}$$

the logarithm of the weighted power mean of $(\alpha, 1-\alpha)$ of order $1/\kappa$ with weights $(\beta, 1-\beta)$. Welfare per voter is the sum of private content utility and the non-rival accountability payoff,

$$W(\kappa;F) \;=\; \mathbb{E}_F\big[v(\alpha;\kappa,\beta)\big] \;+\; g\big(\tilde{t}_n(\kappa;F)\big), \qquad \tilde{t}_n(\kappa;F) = \mathbb{E}_F\big[t_n^*(\alpha;\kappa,\beta)\big]. \tag{30}$$

Here v is the value of the voter's content problem alone — the maximum of content value net of steering cost — so the public good enters (30) only through the second term. By the normalization of Section 3.2, both terms are measured in units of content utility.

The public good g is derived from the incumbent's problem rather than assumed. Lemma 12 in Appendix F shows that it is strictly concave for all parameter values, and strictly increasing whenever $\tau_2 > \underline{\tau}\,\tau_1$, for instance whenever $\tau_2 \geq \tau_1$. Section 7 explains why g keeps rising even though, at high scrutiny, the election becomes worse at distinguishing good incumbents from bad ones.

Writing $\tilde{t}_n^B = \tilde{t}_n(\kappa_B;F)$ for the average diet in the bundled environment, moving to the unbundled environment changes welfare by

$$\Delta W \;=\; \underbrace{G(F)}_{\text{private gain}} \;+\; \underbrace{g\big(\tilde{t}_n^B + \Delta(F)\big) - g\big(\tilde{t}_n^B\big)}_{\text{public effect}}, \qquad G(F) \equiv \mathbb{E}_F\big[v(\alpha;\kappa_U,\beta) - v(\alpha;\kappa_B,\beta)\big]. \tag{31}$$

From now on, G, Δ and the critical stake S introduced in Proposition 7(iii) are evaluated in the large-electorate limit, where the civic term $\zeta = O(1/N)$ vanishes and the two environments differ only through κ. Voters do internalize the public good when choosing their diets — that is the term $\zeta = g'(\tilde{t}_n)/(N\kappa)$ in (21) — but only at rate $1/N$, so in the limit v is exactly the utility they attain.[7]

Proposition 7 (Welfare effects of unbundling)**.** *Let $\tau_2 > \underline{\tau}\,\tau_1$ (for instance $\tau_2 \geq \tau_1$), so that g is strictly increasing and strictly concave on $[0,1]$ (Lemma 12), and suppose news is produced in both environments. Unless stated otherwise, the statements hold for any default bundle β.*

(i) ***Private Pareto improvement.*** *For every voter with $\alpha \neq \frac{1}{2}$, $v(\alpha;\kappa_U) > v(\alpha;\kappa_B)$; voters with $\alpha = \frac{1}{2}$ are unaffected. Hence $G(F) > 0$ for every F other than the point mass at $\frac{1}{2}$. If $\beta = \frac{1}{2}$ the gain is symmetric, $v(1-\alpha;\kappa) = v(\alpha;\kappa)$.*

(ii) ***Electorates whose scrutiny does not fall always gain.*** *If $\Delta(F) \geq 0$, then $\Delta W > 0$. When $\beta = \frac{1}{2}$, this holds for every F symmetric about $\frac{1}{2}$, for which $\Delta(F) = 0$ and $\Delta W = G(F)$, and in the Beta family for every $\mu \geq \frac{1}{2}$.*

(iii) ***Threshold on the political stake.*** *Suppose $\Delta(F) < 0$. Unbundling reduces welfare if and only if the loss of accountability exceeds the private gain,*

$$g\big(\tilde{t}_n^B\big) - g\big(\tilde{t}_n^B + \Delta(F)\big) > G(F).$$

[7] In an electorate of finite size, the civic term is strictly positive, so a voter consumes slightly more news than the content-only optimum at which v is evaluated. Because he was at an optimum, this deviation costs him only an amount of second order in ζ, that is $O(1/N^2)$.

Equivalently, defining the critical stake

$$S(F) \;\equiv\; \frac{G(F)}{|\Delta(F)|} \;\in\; (0,\infty),$$

welfare falls if $g'(\tilde{t}_n^B) > S(F)$*, and only if* $g'\big(\tilde{t}_n^B + \Delta(F)\big) > S(F)$.

Proof in Appendix F.

The proposition assumes that news remains viable after unbundling. If unbundling instead pushes demand below the critical mass (Proposition 2), voters are rationed to $t_n = 0$ and lose content value as well as scrutiny. Part (i) then fails for news-leaning voters, and the welfare loss exceeds the one computed here.

Proposition 7 has the following consequences.

Everyone benefits from unbundling privately. By part (i), every voter's content utility is strictly higher in the unbundled environment, so the transition faces no private opposition — and yet, by part (iii), it can lower welfare. This is the model's central externality. Each voter internalizes the full gain from a diet closer to his taste, but only $1/N$ of the effect of his attention on accountability, so the individually optimal response to cheaper steering is collectively excessive. Because the public good falls for everyone, a voter whose private gain is small can be worse off overall even though his content utility has risen.

Homogeneity of preferences can aggravate the welfare loss. At the benchmark technologies, concentrating tastes around a given mean shrinks the private gain: numerically, for every default examined, the gain $v(\alpha;\kappa_U) - v(\alpha;\kappa_B)$ is convex in α, so by Jensen's inequality any mean-preserving spread raises $G(F)$, whatever the electorate. This convexity is not general; it fails, for instance, for $(\kappa_B, \kappa_U) = (1, \frac{1}{2})$. The effect of concentration on the loss of scrutiny holds more narrowly still. With an evenly mixed default and $\kappa_B \geq 1 \geq \kappa_U$, concentration deepens that loss when no voter leans towards news (Proposition 6(iv)) and, numerically, within the Beta family at the benchmark technologies. Where both effects hold, the two terms of (31) move the same way, so the critical stake $S(F)$ falls as the electorate becomes more homogeneous, as Figure 4 shows for the Beta family at the benchmark calibration. For other taste distributions or technologies, a more homogeneous electorate need not have a lower critical stake.

What results depend on the size of the public good? Parts (i) and (ii) do not depend on how the public good is scaled: every voter gains privately, and electorates whose scrutiny does not fall gain in welfare, whatever the values of (τ_1, τ_2). Nor does the critical stake $S(F)$, which is built from tastes and technology alone, or the way it varies across electorates. What does depend on the size of the public good is whether a particular electorate ends up on the harmful side of the threshold. The scale is summarized by $g(1) - g(0)$, the value to a voter of moving from no scrutiny at all to full scrutiny; since g is linear in (τ_1, τ_2), scaling both by a common factor scales $g(1) - g(0)$, and every derivative of g, by the same factor. In the reduced form g is close to linear over the relevant range, so the comparison in part (iii) is, to a good approximation, a comparison between $S(F)$ and $g(1) - g(0)$: unbundling lowers welfare when the value of full scrutiny exceeds the electorate's critical stake.

Figure 4(a) shows the private Pareto improvement: every voter gains except those with exactly balanced tastes, and the largest gains accrue to voters with extreme tastes. Panel (b)

shows that welfare can nonetheless fall. It plots the critical stake $S(F) = G(F)/|\Delta(F)|$ for Beta electorates with three levels of concentration. Welfare falls where $S(F)$ lies below the marginal value of scrutiny. The horizontal line marks $g' \approx 0.87$, its value at the bundled scrutiny of the benchmark electorate, and the bar beneath shows the exact range of average tastes over which unbundling lowers welfare at these stakes. At the benchmark calibration, and holding average interest in news fixed, a Beta electorate whose tastes are not sufficiently dispersed loses more in accountability than it gains in content.

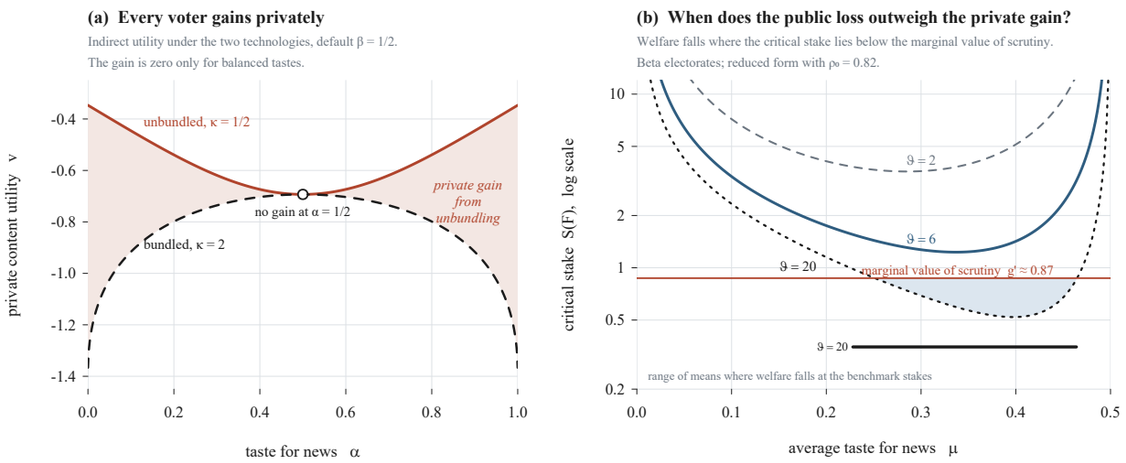


Figure 4: Welfare effects of unbundling. Panel (a): private content utility under the two technologies; the shaded region is the private gain, strictly positive except at $\alpha = \frac{1}{2}$. Panel (b): the critical stake $S(F)$ for Beta electorates with $\vartheta \in \{2, 6, 20\}$, the marginal value of scrutiny g' at the bundled scrutiny of the benchmark electorate, and the range of average tastes over which welfare falls. Parameters: $\tau_1 = 1$, $\tau_2 = 6$, $\gamma = 0.6$, $\rho_0 = 0.82$, $\kappa_B = 2$, $\kappa_U = \frac{1}{2}$, $\beta = \frac{1}{2}$; reduced form of Section 4.1.

The welfare results use only two properties of the public good: it rises with scrutiny, and at a decreasing rate. Section 7 turns to the mechanism behind them, through the incumbent's effort and the voters' choice of whom to re-elect.

7 Discipline and selection

Scrutiny always raises the effort of a bad incumbent, and discipline improves monotonically (14). What is not monotone is how well the election separates good incumbents from bad ones. Voters never observe the incumbent's type: they see at most a verdict, and a bad incumbent can pass. I measure this by the *selection gap*, the difference between the probabilities that a good and a bad incumbent are re-elected. By (12) and (15),

$$\rho_g - \mathbb{E}_c[\rho_b] = \tilde{t}_n - \frac{\tilde{t}_n^2}{2}\Big(1 - \ln\frac{\tilde{t}_n}{2}\Big). \tag{32}$$

The mechanism can be rewritten as

$$\rho_g - \mathbb{E}_c[\rho_b] = \tilde{t}_n\Big(1 - \mathbb{E}_c\big[\sqrt{e_1^*}\big]\Big). \tag{33}$$

The gap is the product of the share of voters who are informed, which rises with scrutiny, and the extent to which a pass distinguishes a good incumbent from a bad one, which falls as bad incumbents respond by working harder. The first element is stronger at low scrutiny

and the second at high scrutiny. The uninformed voters' contribution, $(1-\tilde{t}_n)\rho_0$, is the same for both types and cancels. The derivative of the gap, $1-\tilde{t}_n/2+\tilde{t}_n\ln(\tilde{t}_n/2)$, equals one at $\tilde{t}_n=0$ and $\frac{1}{2}-\ln 2<0$ at $\tilde{t}_n=1$, and its second derivative, $\frac{1}{2}+\ln(\tilde{t}_n/2)$, is negative in the interval $[0,1]$. The gap is therefore strictly concave, with a unique interior maximum at $\tilde{t}_n^*\approx 0.569$, where it equals 0.204, against 0.153 at full attention. These closed forms are those of the reduced form. When the voting environment is held fixed, the selection gap keeps its shape but peaks at higher scrutiny, and below full scrutiny only if the electoral reward is strong enough (Appendix A). Selection is sharpest at intermediate scrutiny and deteriorates when attention becomes very high — not because voters learn less, but because bad incumbents respond by working harder.

Figure 5(a) plots the probabilities of re-election. A good incumbent is re-elected with probability (12), which rises linearly from ρ_0 to one. The solid red curve is (15), a bad incumbent's probability of re-election averaged over her cost. The dashed curve shows a single bad incumbent with cost $c=0.5$: by (13) she exerts effort $\tilde{t}_n^2$, so she is re-elected with probability $\tilde{t}_n^2+(1-\tilde{t}_n)\rho_0$, and she reaches full effort only when every voter is informed. The shaded area is the selection gap. At low scrutiny a bad incumbent's expected probability of re-election falls, with slope $-\rho_0$ at $\tilde{t}_n=0$: the first voters to become informed mostly observe a fail, before her effort responds. Panel (b) plots the gap (32), which does not depend on ρ_0.

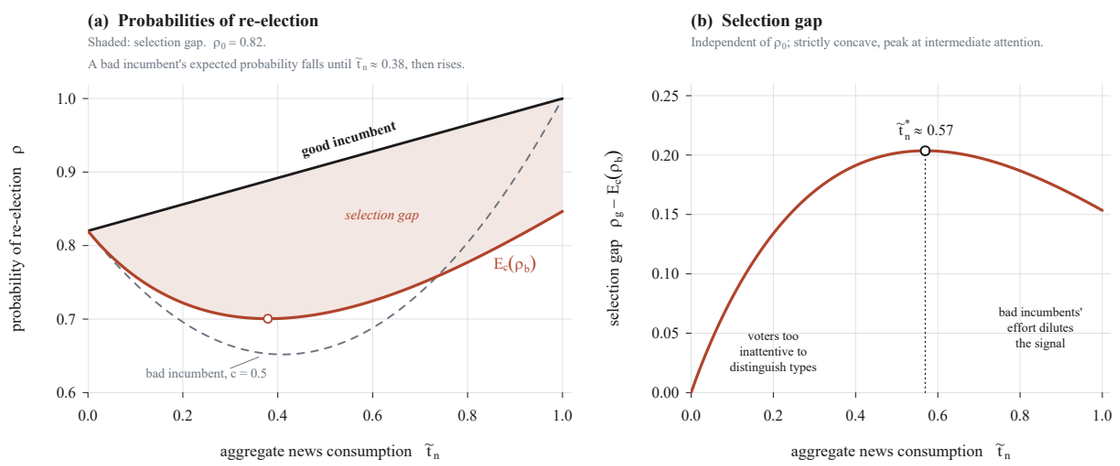


Figure 5: Probabilities of re-election and the selection gap. Parameters: $\rho_0=0.82$, $c\sim U(0,1)$.

The fixed-cost curve isolates the strategic behavior of an individual bad type. If $c<\tilde{t}_n/2$, the incumbent exerts full effort and becomes observationally indistinguishable from a good type. Their cost of exerting effort is relatively low to the level of monitoring, and heightened scrutiny makes imitation of the good type optimal. As $\tilde{t}_n$ rises, the set of bad types for whom this is true becomes larger, and when $\tilde{t}_n=1$ it consists of half of them. Hence, in this context, better discipline does not necessarily mean better selection of politicians. Greater attention always raises effort, so a bad incumbent under intense scrutiny delivers more first-period public good than one under weak scrutiny. However, it also makes her harder to tell apart from a good type. As a result, the electorate is more likely to re-elect a bad politician who was imitating. When monitoring is weak, voters are too inattentive to distinguish types at all. When it is very strong, good and bad types are harder to distinguish as the latter imitates the former. Electoral selection is strongest in an intermediate range — a "Goldilocks" region of voter attention. This mechanism is consistent with Prato and

Wolton (2016), who show that excessive voter interest can encourage low-quality politicians to pander.

The single-peaked selection gap does not make the public good single-peaked. A good incumbent is re-elected more often at every level of scrutiny, but the officeholder in the second period is also good when a bad incumbent is replaced by a challenger, who is good with probability γ, and re-electing more bad incumbents forgoes this. The probability that the second-period officeholder is good is $\gamma + \gamma(1-\gamma)\big(\rho_g - \mathbb{E}_c[\rho_b]\big)$, which moves with the selection gap. Once demand for news is above $\tilde{t}_n^*$, it is less likely to have a good officeholder in the second period. The second-period public good can be written as

$$\gamma\tau_1 + \gamma(1-\gamma)\tau_1\big(\rho_g - \mathbb{E}_c[\rho_b]\big) + \gamma(\tau_2 - \tau_1)\,\rho_g, \tag{34}$$

where the second term is the selection effect and the third is the value of keeping an experienced good incumbent rather than replacing her. Two other forces offset the loss from weaker selection: bad incumbents work harder in the first period, and good incumbents are re-elected more often, which raises the third term whenever $\tau_2 > \tau_1$ (Lemma 12). The "Goldilocks" region is therefore a statement about who holds office in the second period, not about welfare: excessive monitoring worsens selection, but the public good still rises.

Corollary 8 (Political consequences of unbundling)**.** *Suppose news is produced in both environments and unbundling lowers aggregate scrutiny from $\tilde{t}_n$ to $\tilde{t}_n - |\Delta|$, and let $\tilde{t}_n^* \approx 0.57$ be the level of scrutiny at which the selection gap (32) peaks. Then:*

(i) ***Discipline weakens:*** *a bad incumbent's expected effort falls by $|\Delta|\big(1 - \frac{\tilde{t}_n}{2}\big) + \frac{|\Delta|^2}{4}$.*

(ii) ***Good incumbents are re-elected less often:*** *ρ_g falls by $|\Delta|(1-\rho_0)$.*

(iii) ***The public good falls:*** *$g(\tilde{t}_n - |\Delta|) < g(\tilde{t}_n)$ whenever g is increasing (Lemma 12).*

(iv) ***Selection may improve or worsen:*** *the gap narrows if $\tilde{t}_n \le \tilde{t}_n^*$, widens if $\tilde{t}_n - |\Delta| \ge \tilde{t}_n^*$, and may move either way if the fall in scrutiny crosses $\tilde{t}_n^*$.*

Proof. (i) By (14), $\mathbb{E}_c[e_1^*](\tilde{t}_n) - \mathbb{E}_c[e_1^*](\tilde{t}_n - |\Delta|) = |\Delta|\big(1 - \frac{\tilde{t}_n}{2}\big) + \frac{|\Delta|^2}{4}$. (ii) By (12), ρ_g is linear in $\tilde{t}_n$ with slope $1-\rho_0$. (iii) is Lemma 12. (iv) The gap (32) is strictly concave with its maximum at $\tilde{t}_n^*$, so it is increasing below $\tilde{t}_n^*$ and decreasing above. If $\tilde{t}_n - |\Delta| < \tilde{t}_n^* < \tilde{t}_n$, the two levels lie on opposite sides of the peak, and the gap widens if and only if its value at $\tilde{t}_n - |\Delta|$ exceeds its value at $\tilde{t}_n$. □

Lower scrutiny always weakens incentives: bad incumbents work less, and good incumbents are re-elected less often because fewer voters are informed. Its effect on selection depends on the initial level of attention. Where attention is modest, as in the benchmark, lower scrutiny also makes good and bad incumbents harder to tell apart, so discipline and selection weaken together. Only where attention is very high does lower scrutiny improve selection, because fewer bad incumbents then find it worth imitating good ones. Even in that case the public good falls, because g is increasing (Lemma 12).

8 Conclusion

This paper asks what happens to political accountability when exposure to political information stops being incidental and becomes a matter of choice. Modeling media technology

as the price of steering attention away from whatever bundle a media platform supplies by default, I show that lowering that price can reduce aggregate scrutiny and weaken an incumbent's incentive to exert effort — while making every individual voter better off.

The argument runs in three steps. Unbundling redistributes attention: entertainment-leaning voters withdraw from news and news-leaning voters specialize into it, with the biggest changes at intermediate tastes. With an evenly mixed default (news and entertainment supplied in the same proportions), those responses are mirror images and cancel in a balanced electorate, so aggregate scrutiny moves only with the asymmetry of tastes: it falls whenever average interest in news lies below the midpoint. A news-heavy default moves this threshold up and an entertainment-heavy one moves it down. Aggregate scrutiny falls the most in an electorate whose voters all share the same moderate preference for entertainment. Aggregate scrutiny has two implications for elections: it always raises incumbent effort, but it also makes types harder to tell apart, as low-cost bad incumbents imitate good ones. Moreover, every voter gains privately from the change, so the whole welfare cost falls on a public good that each voter internalizes only in small part.

Note that in this model there is no misinformation: news is accurate and freely available wherever it is produced. There is no manipulation: media firms are demand-followers, with no power over the composition of what voters consume and no influence beyond the discrete decision of whether reporting exists at all. There is no irrationality: voters correctly anticipate the incumbent's effort and optimize given it. And there is no sorting into ideological camps: tastes here are over content type, not over political position, and the model contains no partisanship. Nevertheless, political accountability weakens.

These findings rationalize the contemporary media landscape. According to the 2026 Reuters Institute Digital News Report, news consumption via traditional channels such as TV, newspapers, and "owned and operated" news websites has been declining for years across 48 surveyed countries. At the same time, the interest in news has also been declining: since 2021, the proportion of people who are interested in news has fallen by 13 percentage points (Egan et al. 2026). Two trends have thus coincided: news has become easier to avoid, and fewer people want it. In the model, unbundling lowers scrutiny only where voters, on average, prefer entertainment, so falling interest in news makes the effect of cheaper steering on the loss of accountability stronger. This has significant consequences for local markets. Investigative journalism is costlier than its potential substitution (reprinted wire stories, listings, weather), and with the internet supplying that content directly, many localities risk falling below the critical mass of attention that sustains a newsroom at all.[8]

In this context, policy implications focus on the demand for news rather than on its supply. Interventions that lower the cost of producing news relax the critical mass and can restore local provision, but they do not change how much news voters choose to consume once it exists. Instead, measures that raise the value voters place on being informed would improve accountability in this model. They matter most where unbundling mostly affects welfare: in small constituencies, where the critical mass of attention is hardest to reach, and where voters on average prefer entertainment.

There are several potential extensions to this model. The default bundle is exogenous here,

[8] Local news: How publishers can still provide value in a platform world by Richard Fletcher, June 2025. Accessed 7 September 2026.

and endogenizing could be a next step. The interesting extension would combine the choice of default with a participation decision. In this setting, consumers can abandon a media firm whose default bundle is not close enough to their preferred choice of content. In turn, media platforms can personalize the default mix similarly to recommender systems. Moreover, if the taste for news depended on past consumption, a feedback loop, in which unbundling lowers news consumption, which lowers the taste for news, which lowers consumption again, would close. Third, Appendix B lets voters differ in how much they value the public good, but this valuation is independent of the taste for media content. Correlating the two would make the two channels add up rather than average out. This is plausible, as those who prefer entertainment are also likely to care less about local government.

The comparative statics presented in the paper are, in principle, measurable with data. There are two parameters for taste: the mean μ and the concentration ϑ. The relevant parameter for the media environment is the default bundle β. With an evenly mixed default, unbundling lowers average demand for news if and only if $\mu < \frac{1}{2}$ (Proposition 6(iii)). A news-heavy default raises this threshold, and an entertainment-heavy one lowers it. When the bundled technology compresses tastes, and the unbundled one amplifies them, the fall is larger in an electorate whose voters all share the same taste than in one whose tastes are spread below the midpoint (Proposition 6(iv)). Neither prediction requires tastes to be observed directly, as they can be recovered from media diets by inverting (21). Testing them with local government data would be a natural empirical companion to this paper.

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A Micro-foundation: probabilistic voting

This appendix derives the probability of re-election in Section 4.1 from the voting rule of Section 3.5.

Voters fall into three groups, $i \in \{\text{pass}, \text{fail}, U\}$, with the beliefs (11) and benefits B_i of Section 4.1. Since each voter learns a verdict with probability equal to his news share, independently across voters, a share $\tilde{t}_n$ of voters is informed in a large electorate.

Assumptions. Voter J's idiosyncratic assessment is $\psi_J \sim U[-A, A]$, independent across voters and of verdicts, and the common popularity shock is $\xi \sim U[a, b]$. The following two conditions ensure that neither distribution is truncated, so that probabilities are linear in benefits. First:

$$B_i + \xi \in [-A, A] \tag{35}$$

A is large enough that for every group and every realization of ξ: no group votes unanimously, and each group's vote share is linear in its benefit (Step 1). Second:

$$\begin{aligned} a &\leq -B_{\text{pass}} \\ b &\geq -B_{\text{fail}} \end{aligned} \tag{36}$$

The popularity shock is wide enough that any change in voters' average benefit changes the probability of re-election. As Step 2 shows, the incumbent is re-elected when voters' average benefit from keeping her, plus the shock, is positive. That average lies between B_{fail} and B_{pass}, and the conditions ensure that, for every value in this range, some shocks re-elect her and others do not, so the probability of re-election is linear in the average benefit.

Step 1: vote shares. Fix ξ. Voter J's gain from re-electing the incumbent is $B_i + \psi_J + \xi$: the expected benefit for his group, plus his idiosyncratic assessment of the candidates, which captures considerations other than performance such as party loyalty or personal appeal, plus the common shock. He votes for her if this gain is non-negative, that is, if $\psi_J \geq -(B_i + \xi)$. Even a voter who saw a fail, and so knows she is bad, may vote for her if his idiosyncratic assessment is favorable enough. By the condition (35), the share of his group doing so is

$$\Pr\big(\psi_J \geq -(B_i + \xi)\big) = \frac{1}{2} + \frac{B_i + \xi}{2A}. \tag{37}$$

For a bad incumbent who exerted e_1, the uninformed make up $1 - \tilde{t}_n$ of voters, those who saw a pass $\tilde{t}_n\, p(e_1)$, and those who saw a fail $\tilde{t}_n\big(1 - p(e_1)\big)$. Weighting by these shares, which sum to one,

$$VS = \frac{1}{2} + \frac{\tilde{t}_n\, B_I(e_1) + (1 - \tilde{t}_n)\, B_U + \xi}{2A}, \tag{38}$$

with B_I as in Section 4.1. For a good incumbent, every informed voter sees a pass, which is the case $p = 1$.

Step 2: majority rule. The incumbent is re-elected if $VS > \frac{1}{2}$, that is if $X + \xi > 0$ with $X \equiv \tilde{t}_n B_I(e_1) + (1 - \tilde{t}_n) B_U$. Since ξ is continuous, ties occur with probability zero. Because X lies between B_{fail} and B_{pass}, the condition (36) gives $-X \in [a, b]$, so

$$\rho = \Pr(\xi > -X) = \Gamma(X), \qquad \Gamma(X) \equiv \frac{X + b}{b - a}, \tag{39}$$

which is affine in X.

Step 3. Because Γ is affine and X is a weighted average of B_I and B_U with weights summing to one,

$$\rho = \tilde{t}_n\,\Gamma\big(B_I(e_1)\big) + (1-\tilde{t}_n)\,\Gamma(B_U) \equiv \tilde{t}_n\,q(e_1) + (1-\tilde{t}_n)\,\rho_0, \tag{40}$$

and a good incumbent is re-elected with probability $\rho_g = \tilde{t}_n\,q(1) + (1-\tilde{t}_n)\rho_0$. Since B_I is affine in $p(e_1)$ with slope $B_{\text{pass}} - B_{\text{fail}} = \pi_{\text{pass}}\tau_2 > 0$,

$$q'(e_1) = \frac{\pi_{\text{pass}}\tau_2}{b-a}\,p'(e_1) > 0, \qquad q''(e_1) = \frac{\pi_{\text{pass}}\tau_2}{b-a}\,p''(e_1) \le 0, \tag{41}$$

with strict concavity whenever p is strictly concave. Since Γ is increasing and $B_U > B_{\text{fail}}$, $\rho_0 = \Gamma(B_U) > \Gamma(B_{\text{fail}}) = q(0) \ge 0$.

The normalization. The main text sets $q(e_1) = \sqrt{e_1}$. This requires $p(e_1) = \sqrt{e_1}$, $q(0) = \Gamma(B_{\text{fail}}) = 0$ and a unit slope, $\pi_{\text{pass}}\tau_2 = b - a$; both conditions (35) and (36) then hold with equality, $b = -B_{\text{fail}}$ and $a = -B_{\text{pass}}$. The probability that an uninformed voter's group re-elects the incumbent is thus:

$$\rho_0 = \Gamma(B_U) = \frac{B_U - B_{\text{fail}}}{B_{\text{pass}} - B_{\text{fail}}} = \frac{\gamma}{\pi_{\text{pass}}} = \gamma + (1-\gamma)\,\bar{p}_b \;\ge\; \gamma. \tag{42}$$

The derivation holds the voting environment (A, a, b) fixed. In that environment ρ_0 is constant, because uninformed voters rely on the prior alone, while the slope of q varies with scrutiny through π_{pass}, which depends on the effort voters expect from bad incumbents. Normalizing q to $\sqrt{e_1}$ at every level of scrutiny would instead require (a, b) to vary with scrutiny, and $\rho_0 = \gamma + (1-\gamma)\bar{p}_b$ would vary with it. The microfoundation therefore rationalizes three features of (1): the weighting of informed and uninformed voters by news consumption, the fact that only informed voters respond to effort, and the restriction $\rho_0 > q(0)$, which becomes $\rho_0 \ge \gamma$ under the normalization. It does not rationalize the invariance of q and ρ_0 across $\tilde{t}_n$.

B Extension: heterogeneous valuation of the public good

The main text places weight one on the public good for every voter. This appendix relaxes this restriction: voters value the public good at $\lambda_J > 0$. The main text is the special case $\lambda_J \equiv 1$: a voter with $\lambda_J < 1$ values governance quality less than the baseline citizen; one with $\lambda_J > 1$ values it more. I write $\bar{\lambda} = \frac{1}{N}\sum_J \lambda_J$ for the population mean, so that $\bar{\lambda} \gtrless 1$ says whether the electorate values governance more or less than the main text assumes. Here, I assume that $\beta = \frac{1}{2}$.

Let voter J place private valuation $\lambda_J > 0$ on the public good, so that his objective becomes

$$\max_{t\in(0,1)}\; t\ln\alpha_J + (1-t)\ln(1-\alpha_J) - C_\kappa(t) \;+\; \lambda_J\,g(\tilde{t}_n), \qquad \tilde{t}_n = \tfrac{1}{N}\textstyle\sum_I t_I. \tag{43}$$

The public good remains nonrival — every voter enjoys $g(\tilde{t}_n)$ undivided — and a voter's own attention influences the aggregate only at the rate $1/N$. Differentiating (43) and using $C'_\kappa(t) = \kappa\ln\frac{t}{1-t}$, the first-order condition is

$$\kappa\ln\frac{t}{1-t} \;=\; x_J + \frac{\lambda_J\,g'(\tilde{t}_n)}{N}, \qquad x_J = \ln\frac{\alpha_J}{1-\alpha_J},$$

which inverts to

$$t^*(\alpha_J, \lambda_J; \kappa) = \frac{\alpha_J^{1/\kappa}\, e^{\zeta_J}}{\alpha_J^{1/\kappa}\, e^{\zeta_J} + (1-\alpha_J)^{1/\kappa}}, \qquad \boxed{\zeta_J \;\equiv\; \frac{\lambda_J\, g'(\tilde{t}_n)}{N\kappa}} \tag{44}$$

where ζ_J is voter J's *civic term*.

Two features of (44) make the analysis distinct from the model from the main text:

Remark 9. The civic term $\zeta_J = \lambda_J g'(\tilde{t}_n)/(N\kappa)$ is decreasing in the price of steering κ. Civic concern, therefore, translates into news consumption more strongly in unbundled environments: for given λ_J, halving κ doubles the term.

Unbundling lets both tastes α_J and λ_J express themselves. A society with a strong civic culture is correspondingly less exposed to the accountability cost of unbundling.

Proposition 10 (Two roles of the civic valuation)**.** *The civic valuation λ_J enters the model twice, at different orders:*

(i) ***Behavior (order*** $1/N$***).*** *Voter J's civic motive raises the log-odds of his diet by $\zeta_J = \lambda_J g'(\tilde{t}_n)/(N\kappa)$, and his news share by at most $\zeta_J/4$. The effect vanishes as $N \to \infty$.*

(ii) ***Welfare (order*** 1***).*** *The social cost per voter of a fall Δ in scrutiny is $\bar{\lambda}\big[g(\tilde{t}_n) - g(\tilde{t}_n - \Delta)\big]$, which does not vanish with N.*

Hence a change in civic valuations alters what unbundling is worth far more than what citizens do about it.

Proof. (i) By (44), the log-odds of voter J's diet are $x_J/\kappa + \zeta_J$, so the civic motive adds ζ_J to them. Since the slope of Λ is at most $\frac{1}{4}$, his news share rises by at most $\zeta_J/4 \le \lambda_J \bar{g}/(4N\kappa)$, where $\bar{g} = g'(0)$ bounds g' (Lemma 12); this vanishes as $N \to \infty$. (ii) Welfare per voter is $\frac{1}{N}\sum_J \big[v_J + \lambda_J g(\tilde{t}_n)\big]$, whose public-good term is $\bar{\lambda}\, g(\tilde{t}_n)$. A fall Δ in scrutiny therefore lowers it by $\bar{\lambda}\big[g(\tilde{t}_n) - g(\tilde{t}_n - \Delta)\big]$, which does not depend on N. □

This asymmetry is the free-rider problem. Citizens who value governance twice as much consume only marginally more news, because their attention increases aggregate scrutiny only at the rate $1/N$. However, they suffer twice the loss when scrutiny falls. Therefore, civic valuation is relevant mostly for welfare, not for behavior. In this model, a society cannot count on citizens caring more about public good to sustain the optimal level of scrutiny.

C Equilibrium

Additional notation. Let $N \in \mathbb{N}$ denote the number of voters. Voter J has taste $\alpha_J \in (0,1)$ and, given the media environment (κ, β) and aggregate scrutiny, a desired news share $t^*_{J,n} \in (0,1)$. Let $\nu \in \{0,1\}$ indicate whether news is produced. Since news is supplied only by the modeled firms, and is unconstrained where a newsroom exists, realized consumption is

$$\tilde{t}_{J,n} := \nu\, t^*_{J,n}, \qquad \tilde{t}_{J,e} := 1 - \tilde{t}_{J,n}, \tag{45}$$

and average desired and realized news consumption are

$$t_n^* := \frac{1}{N}\sum_{J=1}^{N} t_{J,n}^*, \qquad \tilde{t}_n := \frac{1}{N}\sum_{J=1}^{N} \tilde{t}_{J,n} = \nu\, t_n^*. \tag{46}$$

Denote $x_J \equiv \ln \frac{\alpha_J}{1-\alpha_J}$ for voter J's taste log-odds and $\bar{b} \equiv \ln \frac{\beta}{1-\beta}$ for the log-odds of the environment's default mix, so that the marginal steering cost is $C'_{\kappa,\beta}(t_{J,n}) = \kappa\big[\ln \frac{t_{J,n}}{1-t_{J,n}} - \bar{b}\big]$.

Media profits. Advertising revenue is proportional to attention devoted to each content type on the platform, at rates $A_n > A_e > 0$. A platform pays a fixed cost FC_e to operate and, if it funds a newsroom, an additional fixed cost FC_n — while news carries no marginal cost once the newsroom exists. With $M \in \mathbb{N}$ symmetric platforms sharing the market, per-firm profit given the newsroom decision ν is

$$\Pi(\nu; d, M) = \frac{N}{M}\Big[A_n\, \nu d + A_e\big(1 - \nu d\big)\Big] - FC_e - \nu\, FC_n, \tag{47}$$

or equivalently

$$\Pi(\nu; d, M) = \frac{N}{M}\Big[A_e + \nu\,(A_n - A_e)\, d\Big] - FC_e - \nu\, FC_n. \tag{48}$$

Entertainment revenue A_e is gained on all attention regardless of ν. Funding a newsroom gives the premium $A_n - A_e$ on the share of attention devoted to the news, d.

Definition. An equilibrium is a profile

$$\Big((t_{J,n}^*)_{J=1}^N,\ \big(e_1^\theta, e_2^\theta\big)_{\theta\in\{g,b\}},\ \nu,\ M,\ \tilde{t}_n,\ \rho\Big) \tag{49}$$

such that:

(i) **Consumers.** Each voter J chooses $t_{J,n}^* \in (0,1)$ to maximize content value net of steering cost, internalizing the effect of his own attention on the public good:

$$t_{J,n}^* \in \arg\max_{t_{J,n}\in(0,1)} \Big[t_{J,n} \ln \alpha_J + (1 - t_{J,n}) \ln(1-\alpha_J)\Big] - C_{\kappa,\beta}(t) + g\Big(\tfrac{t_{J,n} + \sum_{I\neq J} \tilde{t}_{I,n}}{N}\Big), \tag{50}$$

taking the diets of others as given. Equivalently, $t_{J,n}$ satisfies

$$\ln \frac{t_{J,n}}{1 - t_{J,n}} = \frac{x_J}{\kappa} + \bar{b} + \zeta(\tilde{t}_n), \qquad \zeta(\tilde{t}_n) \equiv \frac{g'(\tilde{t}_n)}{N\kappa}, \qquad x_J = \ln \tfrac{\alpha_J}{1-\alpha_J}. \tag{51}$$

(ii) **Media firms (newsroom and entry).** Given the demand a newsroom would attract, $d = t_n^*$, the market has M symmetric platforms with newsroom decision ν such that:

(a) *Viability:* if $M \geq 1$, $\Pi(\nu; d, M) \geq 0$;

(b) *Newsroom optimality:* $\nu = 1$ if and only if $\frac{N}{M}(A_n - A_e)d \geq FC_n$;

(c) *No profitable entry of either format:* $\Pi(0; d, M+1) < 0$ and $\Pi(1; d, M+1) < 0$.

(iii) **Incumbent.** Given realized average news $\tilde{t}_n$, the incumbent chooses effort by backward induction: a good type exerts $e_1^g = e_2^g = 1$; a bad type with cost c exerts $e_2^b = 0$ and $e_1^b(c) = \min\{1, (\tilde{t}_n/2c)^2\}$.

(iv) **Aggregation.** $\tilde{t}_n = \nu\, t_n^*$ with $t_n^* = \frac{1}{N}\sum_J t_{J,n}$.

(v) **Consistency.** Beliefs are Bayesian on the equilibrium path. The marginal public good $g'(\tilde{t}_n)$ entering (51) is evaluated at the incumbent's equilibrium effort from (iii); and $\rho = \rho(\tilde{t}_n, e_1)$.

Existence and uniqueness. Conditions (i) and (iv) are simultaneous: individual diets depend on aggregate scrutiny through the civic term ζ, and aggregate scrutiny is the average of individual diets. Define, for a level of scrutiny $s \in [0,1]$,

$$\Phi(s) \;:=\; \frac{1}{N}\sum_{J=1}^{N} \Lambda\Big(\frac{x_J}{\kappa} + \bar{b} + \zeta(s)\Big), \qquad \Lambda(z) = \frac{1}{1+e^{-z}}, \tag{52}$$

the average diet that voters choose when they expect scrutiny s. An equilibrium with $\nu = 1$ requires $\tilde{t}_n = \Phi(\tilde{t}_n)$.

There is a loop in the model: expected scrutiny changes the marginal value of news, which changes diets, which changes scrutiny. Because g is concave, more scrutiny lowers the marginal value of monitoring. Hence, the loop runs in the stabilizing direction: attention choices are strategic substitutes, and the average diet is a decreasing function of expected scrutiny. A decreasing function crosses the 45° line exactly once, so uniqueness requires no restriction on how strong the feedback is.

Proposition 11. *Assume g is continuously differentiable on $[0,1]$ and twice continuously differentiable on $(0,1]$. Then an equilibrium exists and is unique.*

Proof. Existence and uniqueness of equilibrium scrutiny. Recall from (52) that

$$\Phi(s) = \frac{1}{N}\sum_{J=1}^{N} \Lambda\Big(\frac{x_J}{\kappa} + \bar{b} + \zeta(s)\Big), \qquad \zeta(s) = \frac{g'(s)}{N\kappa}, \qquad \Lambda(z) = \frac{1}{1+e^{-z}}, \tag{53}$$

is the average diet voters choose when they expect scrutiny s. An equilibrium with $\nu = 1$ is a level of scrutiny that reproduces itself, $\tilde{t}_n = \Phi(\tilde{t}_n)$.

Step 1: how the average diet responds to expected scrutiny. Differentiating Φ by the chain rule, and using the logistic identity $\Lambda'(z) = \Lambda(z)\big(1-\Lambda(z)\big)$ — so that the derivative evaluated at voter J's argument is simply $t_{J,n}(1-t_{J,n})$ — gives

$$\Phi'(s) = \underbrace{\frac{g''(s)}{N\kappa}}_{\zeta'(s)} \times \underbrace{\frac{1}{N}\sum_{J=1}^{N} t_{J,n}(1-t_{J,n})}_{R(s)}, \tag{54}$$

the product of two terms. Since $t_{J,n} \in (0,1)$ for every voter, $R(s) \in (0, \frac{1}{4}]$.

Step 2: the sign. Because $R(s) > 0$ and $N\kappa > 0$, the sign of $\Phi'(s)$ is the sign of $g''(s)$. Differentiating the public good twice,

$$g''(t_n) = (1-\gamma)\tau_1\Big[\tfrac{\gamma-1}{2} + \gamma \ln \tfrac{t_n}{2}\Big] < 0 \qquad \text{for all } t_n \in (0,1], \tag{55}$$

Hence $\Phi'(s) < 0$: *the more scrutiny voters expect, the less news each chooses to consume.* The public good is concave in monitoring, so the marginal increase in additional scrutiny matters progressively less. Attention choices are strategic substitutes.

Step 3: exactly one fixed point. Let $\psi(s) := \Phi(s) - s$ measure the gap between the demand for news and the scrutiny voters expected. Then ψ is continuous, and strictly decreasing because $\psi'(s) = \Phi'(s) - 1 < -1 < 0$. At the two ends,

$$\psi(0) = \Phi(0) > 0, \qquad \psi(1) = \Phi(1) - 1 < 0, \tag{56}$$

At the left endpoint, $\psi(0) = \Phi(0) > 0$: even voters who expect no scrutiny at all still consume news for its own sake, since the logistic Λ is strictly positive and the steering cost

rules out a corner diet. At the right endpoint, $\psi(1) = \Phi(1) - 1 < 0$ for the symmetric reason, no voter consuming only news. Since ψ is continuous and changes sign, it has a zero $\tilde{t}_n^\star \in (0,1)$. Since Φ takes values strictly inside $(0,1)$: every voter's diet is interior, so the average is neither 0 nor 1. A continuous, strictly decreasing function that is positive at one end of an interval and negative at the other has exactly one zero. That zero, $\tilde{t}_n^\star$, is the unique fixed point of Φ: the unique level of scrutiny that voters both expect and deliver. Geometrically, Φ is a downward-sloping best-response curve, and a downward-sloping curve crosses the 45° line once and only once.

Step 4: the media market. Let $d = \tilde{t}_n^\star$, the demand a newsroom would attract, and write

$$M_0 = \Big\lfloor \frac{NA_e}{FC_e} \Big\rfloor, \qquad M_1(d) = \Big\lfloor \frac{N\,[A_e + (A_n - A_e)d]}{FC_e + FC_n} \Big\rfloor \tag{57}$$

for the largest numbers of platforms that break even without and with a newsroom. Since profit is strictly decreasing in M, viability and the absence of same-format entry imply $M = M_0$ in an equilibrium without news and $M = M_1$ in one with news. The absence of entry by the other format then requires $M_1 \le M_0$ in the first case, since a newsroom entrant would make $M_0 + 1$ platforms, and $M_0 \le M_1$ in the second. Finally, with M platforms of either format, adding a newsroom raises a platform's revenue by $\frac{N}{M}(A_n - A_e)d$ at cost FC_n, so a platform prefers to report if and only if

$$d \;\ge\; d(M) \equiv \frac{M\,FC_n}{(A_n - A_e)N}. \tag{58}$$

If $M_1 < M_0$, only $(\nu, M) = (0, M_0)$ can be an equilibrium, and it is: at M_0 a newsroom is unprofitable, since $\Pi(1; d, M_0) < 0 \le \Pi(0; d, M_0)$. If $M_1 > M_0$, only $(1, M_1)$ can be, and it is, by the symmetric argument. If $M_1 = M_0 = M \ge 1$, both formats pass the entry tests, and the newsroom condition $d \ge d(M)$ determines ν uniquely; if $M_1 = M_0 = 0$, no platform enters and $\nu = 0$. The media market therefore has exactly one equilibrium.

M_1 is a non-decreasing step function of d, and M_0 does not depend on it. Let

$$d_{\text{lo}} = \frac{M_0(FC_e + FC_n)/N - A_e}{A_n - A_e}, \qquad d_{\text{hi}} = \frac{(M_0 + 1)(FC_e + FC_n)/N - A_e}{A_n - A_e}, \tag{59}$$

so that $M_1 < M_0$ for $d < d_{\text{lo}}$, $M_1 = M_0$ for $d_{\text{lo}} \le d < d_{\text{hi}}$, and $M_1 > M_0$ for $d \ge d_{\text{hi}}$. By the uniqueness argument, news is produced in none of the first region, in all of the third, and in the second exactly when $d \ge d(M_0)$.

Suppose $M_0 \ge 1$. Since $M_0 \le NA_e/FC_e < M_0 + 1$

$$\begin{aligned} (A_n - A_e)\big[d_{\text{lo}} - d(M_0)\big] &= \frac{M_0 FC_e}{N} - A_e \le 0, \\ (A_n - A_e)\big[d_{\text{hi}} - d(M_0)\big] &= \frac{(M_0 + 1)FC_e + FC_n}{N} - A_e > 0, \end{aligned} \tag{60}$$

so $d(M_0)$ lies in the second region, and news is produced if and only if $d \ge d_N \equiv d(M_0)$: the only condition that binds is whether an existing platform finds a newsroom worth its fixed cost. If $M_0 = 0$, the second region has no platform, and news is produced if and only if $d \ge d_N \equiv d_{\text{hi}}$. In both cases d_N depends on N, A_n, A_e, FC_n and FC_e, but not on (κ, β, F), so it is the same under both technologies, and Proposition 2 holds with d_N in place of $\bar{d}$.

It remains to compare d_N with $\bar{d}$. Suppose first that $M_0 \ge 1$. Substituting $M_0 \le NA_e/FC_e$ into $d_N = d(M_0)$ gives $d_N \le \bar{d}$, and substituting $M_0 > NA_e/FC_e - 1$ gives $d_N > \bar{d} -$

$FC_n/[(A_n - A_e)N]$. Hence

$$\bar{d} - \frac{FC_n}{(A_n - A_e)\,N} \;<\; d_N \;\le\; \bar{d}. \tag{61}$$

Integer effects therefore lower the threshold, because platforms earn rents, by at most $FC_n/[(A_n - A_e)N]$, and $d_N \to \bar{d}$ as the market grows. If instead $M_0 = 0$, that is $NA_e < FC_e$, then

$$d_N = d_{\text{hi}} = \frac{(FC_e + FC_n)/N - A_e}{A_n - A_e}, \qquad d_N - \bar{d} = \frac{(FC_e + FC_n)(FC_e - NA_e)}{N\,FC_e\,(A_n - A_e)} > 0 : \tag{62}$$

when entertainment alone cannot sustain a platform, the threshold exceeds $\bar{d}$ and rises as the market shrinks, as in Section 4.3.

The remaining components. News production fixes $\tilde{t}_n = \nu\,\tilde{t}_n^{\star}$. Given $\tilde{t}_n$, the good type's problem is trivial, and the bad type's objective is strictly concave in effort, so her effort is unique; ρ then follows. The equilibrium profile is therefore unique.

□

D Proof of Lemma 4

Denote $x = \ln\frac{\alpha}{1-\alpha}$ and $\bar{b} = \ln\frac{\beta}{1-\beta}$, and set $m \equiv 1/\kappa$. By (21) the optimal diet under technology κ is

$$s_m(\alpha) \;=\; \Lambda\big(mx + \bar{b}\big), \qquad \Lambda(z) = \frac{1}{1 + e^{-z}}, \tag{63}$$

so $t_n^*(\alpha; \kappa, \beta) = s_{1/\kappa}(\alpha)$. Since $\kappa_B > \kappa_U$ we have $m_U \equiv 1/\kappa_U > m_B \equiv 1/\kappa_B$, and $h = s_{m_U} - s_{m_B}$.

I start the proof from Lemma 4(ii):

(ii) Λ is strictly increasing, so $\partial s_m/\partial m = \Lambda'(mx + \bar{b})\,x$ has the sign of x. For $\alpha < \frac{1}{2}$ we have $x < 0$, so s_m is strictly decreasing in m; since $m_U > m_B$ this gives $h < 0$ on $(0, \frac{1}{2})$. For $\alpha > \frac{1}{2}$ we have $x > 0$, the monotonicity reverses, and $h > 0$. Neither step uses the value of $\bar{b}$, so both hold for every default β.

(i) As $\alpha \to 0$, $x \to -\infty$ and $s_m \to 0$ for every $m > 0$; as $\alpha \to 1$, $x \to +\infty$ and $s_m \to 1$. At $\alpha = \frac{1}{2}$ we have $x = 0$, hence $s_m = \Lambda(\bar{b}) = \beta$ for every m: both technologies deliver the default mix, and the gap closes. So h becomes zero at 0, $\frac{1}{2}$ and 1, whatever the default. These are the only zeros, because by (ii) h has a strict sign in each of the two intervals in between.

(iii) In the log-odds parametrization $s_m(\alpha; \beta) = \Lambda(mx + \bar{b})$ with $x = \ln\frac{\alpha}{1-\alpha}$ and $\bar{b} = \ln\frac{\beta}{1-\beta}$, the map $(\alpha, \beta) \mapsto (1-\alpha, 1-\beta)$ sends $x \mapsto -x$ and $\bar{b} \mapsto -\bar{b}$, hence $mx + \bar{b} \mapsto -(mx + \bar{b})$ for every m. Since $\Lambda(-z) = 1 - \Lambda(z)$,

$$s_m(1-\alpha;\, 1-\beta) \;=\; 1 - s_m(\alpha;\, \beta) \qquad \text{for every } m > 0. \tag{64}$$

Differencing this identity across m_U and m_B, the constants cancel and $h(1-\alpha; 1-\beta) = -h(\alpha; \beta)$.

(iv) Let $\beta = \frac{1}{2}$, so $\bar{b} = 0$.

Step 1. With $\beta = \frac{1}{2}$ we have $1 - \beta = \beta$, so part (iii) gives $h(1-\alpha) = -h(\alpha)$ on $(0,1)$. It is enough to work on $(0, \frac{1}{2})$; statements on $(\frac{1}{2}, 1)$ follow by reflection.

Step 2. With $\bar{b} = 0$ write $s_m(\alpha) = \alpha^m / D_m$, $D_m = \alpha^m + (1-\alpha)^m$, and $P = \alpha(1-\alpha)$. Differentiating, the cross-terms cancel and

$$\begin{aligned} s'_m(\alpha) = \frac{m\,P^{m-1}}{D_m^2}, \qquad s''_m(\alpha) = \frac{m\,P^{m-2}}{D_m^3}\,Q_m(\alpha), \\ Q_m(\alpha) = (m-1)(1-2\alpha)\,D_m \;-\; 2mP\big(\alpha^{m-1} - (1-\alpha)^{m-1}\big). \end{aligned} \tag{65}$$

Fix $\alpha \in (0, \frac{1}{2})$, so $1 - 2\alpha > 0$ and $\alpha < 1 - \alpha$; mP^{m-2}/D_m^3 is positive, so $\operatorname{sign} s''_m = \operatorname{sign} Q_m$.

- If $m > 1$: the factor $m - 1$ is positive, so the first term of Q_m is positive; and since the exponent $m - 1$ is positive, $\alpha^{m-1} < (1-\alpha)^{m-1}$, so the bracket is negative and the second term is positive too. Hence $Q_m > 0$ and s_m is strictly convex.
- If $m < 1$: both inequalities reverse. The factor $m - 1$ is negative; and because the exponent $m - 1$ is now negative, then $\alpha^{m-1} > (1-\alpha)^{m-1}$ and the second term is negative as well. Hence $Q_m < 0$ and s_m is strictly concave.
- If $m = 1$: $s_1(\alpha) = \alpha$ and $s''_1 \equiv 0$.

Step 3. The hypothesis $\kappa_B \geq 1 \geq \kappa_U$ is $m_B \leq 1 \leq m_U$, with at least one inequality strict. By Step 2, $s''_{m_U} \geq 0$ and $s''_{m_B} \leq 0$ on $(0, \frac{1}{2})$, and at least one is strict, so

$$h'' = s''_{m_U} - s''_{m_B} > 0 \qquad \text{on } (0, \tfrac{1}{2}) : \tag{66}$$

h is strictly convex there, and strictly concave on $(\frac{1}{2}, 1)$ by Step 1.

Step 4. h is continuous on $[0, \frac{1}{2}]$, strictly convex on the interior, with $h(0) = h(\frac{1}{2}) = 0$ and $h < 0$ in between by (ii). A strictly convex function attains its minimum at a single point, and that point cannot be an endpoint since the endpoint values exceed the interior ones; hence there is a unique interior minimizer $\alpha^\dagger \in (0, \frac{1}{2})$. By Step 1, $1 - \alpha^\dagger$ is the unique maximizer on $(\frac{1}{2}, 1)$. □

E Proof of Proposition 6

Recall that h denotes the technology gap of Lemma 4. By parts (i)–(ii) of that lemma, for every β,

$$h < 0 \text{ on } (0, \tfrac{1}{2}), \qquad h(\tfrac{1}{2}) = 0, \qquad h > 0 \text{ on } (\tfrac{1}{2}, 1), \qquad h(0) = h(1) = 0. \tag{67}$$

Since $t_n^* \in (0,1)$ in both environments, $|h| \leq 1$.

(i) One-sided electorates. Suppose no voter leans towards news, so F is supported on $(0, \frac{1}{2}]$, and that not every voter has taste exactly $\frac{1}{2}$. Then a non-zero share of voters lies strictly below $\frac{1}{2}$, where $h < 0$, while the remaining voters have a taste $\alpha = \frac{1}{2}$, where $h = 0$. Averaging, $\Delta = \mathbb{E}_F[h(\alpha)] < 0$. The case of an electorate in which no voter leans towards entertainment is identical with the signs reversed.

(ii) Balanced electorates. Let F be symmetric, $f(\alpha) = f(1-\alpha)$ for all α.

Step 1: pair up mirror-image voters. Split the integral defining Δ at $\frac{1}{2}$ and substitute α with $1-\alpha$ on the upper half. Using $f(1-\alpha) = f(\alpha)$,

$$\Delta = \int_0^{1/2} \big[h(\alpha) + h(1-\alpha)\big] f(\alpha)\, d\alpha = \int_0^{1/2} H(\alpha)\, f(\alpha)\, d\alpha, \qquad H(\alpha) \equiv h(\alpha) + h(1-\alpha). \tag{68}$$

$H(\alpha)$ is the change in the combined news consumption of a voter with taste α and his mirror image with taste $1-\alpha$. It therefore suffices to show that $H(\alpha)$ has the sign of $\frac{1}{2} - \beta$ at every $\alpha \in (0, \frac{1}{2})$.

Step 2: write the pair's news consumption as a function of the technology. Fix $\alpha \in (0, \frac{1}{2})$, and write $x = \ln \frac{\alpha}{1-\alpha}$, which is negative, and $\bar{b} = \ln \frac{\beta}{1-\beta}$. Let $m = 1/\kappa$. In the large-electorate limit $\zeta = 0$, and (21) gives the log-odds of the two voters' diets:

$$z_1 = mx + \bar{b} \quad \text{(taste } \alpha\text{)}, \qquad z_2 = -mx + \bar{b} \quad \text{(taste } 1-\alpha\text{)}, \tag{69}$$

the second because replacing α by $1-\alpha$ changes the sign of x. Inverting the log-odds, a diet with log-odds z is

$$t_n^* = \Lambda(z) \equiv \frac{1}{1+e^{-z}}, \tag{70}$$

which is (22) written in terms of z.[9] The pair's combined news consumption is therefore

$$\varphi(m) \equiv \Lambda(mx + \bar{b}) + \Lambda(-mx + \bar{b}). \tag{71}$$

Since $H(\alpha) = \varphi(m_U) - \varphi(m_B)$ and $m_U > m_B$, the sign of $H(\alpha)$ is the sign of the slope of (71) which, as Step 5 shows, does not depend on m.

Step 3: differentiate (71). Since $dz_1/dm = x$ and $dz_2/dm = -x$,

$$\varphi'(m) = x\big[\Lambda'(z_1) - \Lambda'(z_2)\big], \qquad \Lambda'(z) = \Lambda(z)\big(1 - \Lambda(z)\big). \tag{72}$$

Raising m moves the two diets in opposite directions; which effect dominates depends on which voter responds more strongly.

Step 4: which voter responds more. Evaluated at a voter's own log-odds, Λ' is the pass-through factor $t_n^*(1-t_n^*)$ of (23). A log-odds of zero is a balanced diet, $t_n^* = \frac{1}{2}$, where pass-through is largest, and pass-through falls as $|z|$ grows. The voter who responds more is the one whose diet has log-odds closer to zero, that is, the one with the smaller $|z|$. Since $|z_1| < |z_2|$ exactly when $z_1^2 < z_2^2$, it is enough to compare squares, which expand without case distinctions:

$$z_1^2 - z_2^2 = \big[(mx)^2 + 2mx\bar{b} + \bar{b}^2\big] - \big[(mx)^2 - 2mx\bar{b} + \bar{b}^2\big] = 4mx\bar{b}. \tag{73}$$

Since $mx < 0$, $z_1^2 < z_2^2$ if and only if $\bar{b} > 0$. Hence $\Lambda'(z_1) > \Lambda'(z_2)$ when $\bar{b} > 0$, $\Lambda'(z_1) < \Lambda'(z_2)$ when $\bar{b} < 0$, and the two are equal when $\bar{b} = 0$.

Step 5: sign the pair's response. Multiplying the bracket in (72) by $x < 0$,

$$\operatorname{sign} \varphi'(m) = -\operatorname{sign} \bar{b} = \operatorname{sign}\big(\tfrac{1}{2} - \beta\big) \tag{74}$$

for every m. Therefore $\operatorname{sign} H(\alpha) = \operatorname{sign}(\frac{1}{2} - \beta)$ at every $\alpha \in (0, \frac{1}{2})$, and since $f \geq 0$, with positive weight on $(0, \frac{1}{2})$ unless F is the point mass at $\frac{1}{2}$, the same sign passes to Δ. In particular $\Delta = 0$ for every symmetric F if $\beta = \frac{1}{2}$, and only then.

[9] Dividing the numerator and denominator of (22) by $\alpha^{1/\kappa}\beta e^{\zeta}$ shows that $t_n^* = \Lambda(z) \equiv 1/(1+e^{-z})$ with $z = x/\kappa + \bar{b} + \zeta$, the right-hand side of (21); in the large-electorate limit $\zeta = 0$, so $z = mx + \bar{b}$.

With a default leaning towards news ($\beta > \frac{1}{2}$, so $\bar{b} > 0$), the member of the pair who prefers entertainment holds the more balanced diet, so unbundling moves him further than it moves his news-leaning counterpart, and the pair's combined news consumption falls. With a news-poor default the roles reverse, and it rises. With an even default the two responses are exactly equal and cancel.

Step 6: reflection. Write $\check{f}(\alpha) = f(1-\alpha)$ for the mirrored electorate. By Lemma 4(iii), $h(1-\alpha; 1-\beta) = -h(\alpha;\beta)$, so substituting α with $1-\alpha$,

$$\Delta(\check{F}; 1-\beta) = \int_0^1 h(\alpha; 1-\beta)\, f(1-\alpha)\, d\alpha = \int_0^1 h(1-\alpha; 1-\beta)\, f(\alpha)\, d\alpha = -\Delta(F;\beta). \quad (75)$$

(iii) Threshold. Write f_μ for the density of Beta$(\vartheta\mu, \vartheta(1-\mu))$ and $\Delta(\mu)$ for the corresponding aggregate effect.

Step 1: limits at the endpoints. Fix $\varepsilon > 0$. Because h is continuous with $h(0) = 0$, there is $\delta > 0$ such that $|h(\alpha)| < \varepsilon/2$ for all $\alpha \le \delta$. Split the electorate at δ. Voters with $\alpha \le \delta$ contribute at most $\varepsilon/2$ in absolute value to Δ. Voters with $\alpha > \delta$ contribute at most their share, since $|h| \le 1$, and that share is small when the mean is small: as $\alpha \ge 0$,

$$\mu = \mathbb{E}[\alpha] \;\ge\; \mathbb{E}\big[\alpha\, \mathbf{1}\{\alpha > \delta\}\big] \;\ge\; \delta\, \Pr(\alpha > \delta), \quad (76)$$

so $\Pr(\alpha > \delta) \le \mu/\delta$. Hence

$$|\Delta(\mu)| \;<\; \frac{\varepsilon}{2} + \frac{\mu}{\delta} \;<\; \varepsilon \qquad \text{whenever } \mu < \varepsilon\delta/2, \quad (77)$$

and $\Delta(\mu) \to 0$ as $\mu \to 0$. Applying the same argument to $1-\alpha$, whose mean is $1-\mu$, and using $h(1) = 0$, gives $\Delta(\mu) \to 0$ as $\mu \to 1$.

Step 2: the likelihood ratio is increasing. For $\mu' > \mu$,

$$r(\alpha) \equiv \frac{f_{\mu'}(\alpha)}{f_\mu(\alpha)} = c\left(\frac{\alpha}{1-\alpha}\right)^{\vartheta(\mu'-\mu)}, \quad (78)$$

with $c > 0$ a constant, and r is strictly increasing in α: raising the mean shifts weight towards voters who like news more.

Step 3: once non-negative, Δ stays positive. Let $\mu' > \mu$ and suppose $\Delta(\mu) \ge 0$. Write $r_0 = r(\frac{1}{2}) > 0$. Then

$$\Delta(\mu') = \int_0^1 h(\alpha)\, r(\alpha)\, f_\mu(\alpha)\, d\alpha = r_0\, \Delta(\mu) + \int_0^1 h(\alpha)\big[r(\alpha) - r_0\big] f_\mu(\alpha)\, d\alpha. \quad (79)$$

The integrand of the last term in (79) is positive at every $\alpha \ne \frac{1}{2}$: below $\frac{1}{2}$ both h and $r - r_0$ are negative, and above $\frac{1}{2}$ both are positive, by (67) and Step 2. Hence the last term is strictly positive, and $\Delta(\mu') > r_0\Delta(\mu) \ge 0$.

Moving weight towards news-leaning voters adds more weight where h is positive than where it is negative, so Δ can only increase once it is no longer negative.

Step 4: at most one threshold. By Step 3, Δ cannot move from a non-negative value back to a negative one as μ increases, and it can equal zero at most once: if $\Delta(\mu_0) = 0$ then $\Delta > 0$ for all $\mu > \mu_0$, and $\Delta < 0$ for all $\mu < \mu_0$, since a non-negative value below μ_0 would force $\Delta(\mu_0) > 0$. So Δ changes sign at most once, from negative to positive, and there is at most one threshold μ^* with $\Delta < 0$ below it and $\Delta > 0$ above it.

Step 5: existence at an even default. Let $\beta = \frac{1}{2}$ and $\mu < \frac{1}{2}$. The Beta density satisfies

$$\frac{f_\mu(\alpha)}{f_\mu(1-\alpha)} = \Big(\frac{\alpha}{1-\alpha}\Big)^{\vartheta(2\mu-1)} > 1 \qquad \text{for } \alpha \in (0, \tfrac{1}{2}), \tag{80}$$

since the base is below one and the exponent is negative. By (28), $\Delta = \int_0^{1/2} h(\alpha)[f_\mu(\alpha) - f_\mu(1-\alpha)]\, d\alpha$, where the bracket is positive and h is negative, so $\Delta(\mu) < 0$. The case $\mu > \frac{1}{2}$ gives $\Delta(\mu) > 0$ by the same argument with the signs reversed, and $\Delta(\frac{1}{2}) = 0$ by part (ii). Hence $\mu^* = \frac{1}{2}$.

Step 6: existence near an even default. Fix ϑ and any $\mu_1 < \frac{1}{2} < \mu_2$. Step 5 gives $\Delta(\mu_1) < 0 < \Delta(\mu_2)$ at $\beta = \frac{1}{2}$. The gap $h(\alpha; \beta)$ is continuous in β and bounded by one, so both $\Delta(\mu_1)$ and $\Delta(\mu_2)$ are continuous in β and keep their signs for all β in some interval $|\beta - \frac{1}{2}| < \varepsilon(\vartheta)$. On that interval Δ takes both signs, and by Step 4 it changes sign exactly once.

(iv) Homogeneity. Let $\beta = \frac{1}{2}$ and $\kappa_B \geq 1 \geq \kappa_U$. By Lemma 4(iv), h is strictly convex on $(0, \frac{1}{2}]$. Since tastes lie in that interval, Jensen's inequality gives

$$\Delta = \mathbb{E}_F[h(\alpha)] \;\geq\; h(\mu),$$

with equality only if every voter has taste μ; and a mean-preserving spread that keeps every taste in $(0, \frac{1}{2}]$, which replaces a voter by two voters straddling his taste within that interval, raises the average of a convex function. Both Δ and $h(\mu)$ are negative, because $h < 0$ on $(0, \frac{1}{2})$. Taking absolute values reverses an inequality between negative numbers — if $a \geq b$ with $a, b < 0$ then $|a| \leq |b|$ — so $|\Delta| \leq |h(\mu)|$, and any mean-preserving spread strictly reduces $|\Delta|$. □

F Proof of Proposition 7

In this proof, G, Δ and S are evaluated in the large-electorate limit, where the civic term $\zeta = O(1/N)$ vanishes and the two environments differ only through κ. News is produced in both environments, as the proposition assumes. Recall the decomposition (31),

$$\Delta W = \underbrace{G(F)}_{\text{private gain}} + \underbrace{g\big(\tilde{t}_n^B + \Delta(F)\big) - g\big(\tilde{t}_n^B\big)}_{\text{public effect}}, \qquad \tilde{t}_n^B = \tilde{t}_n(\kappa_B; F).$$

The proof uses two properties of the public good, that it is increasing and concave in aggregate scrutiny. These are not assumptions: they follow from the model's primitives, as the following lemma shows.

Lemma 12 (Shape of the public good)**.** *Let $\gamma \in (0,1)$, $\rho_0 \in [0,1]$ and $\tau_1 > 0$. Then g defined in (5) is continuously differentiable and strictly concave on $[0,1]$. It is strictly increasing on $[0,1]$ if and only if $g'(1) > 0$; for $\rho_0 < 1$ this is $\tau_2 > \underline{\tau}\,\tau_1$ with*

$$\underline{\tau} = \gamma - \frac{(1-\gamma)\big[\frac{1}{2} - \gamma(\frac{1}{2} + \ln 2) + \gamma\rho_0\big]}{\gamma(1-\rho_0)} < 1, \tag{81}$$

and it holds in particular whenever $\tau_2 \geq \tau_1$.

Proof. Write t for $\tilde{t}_n \in [0,1]$.

Step 1. With $\rho = t\sqrt{e_1} + (1-t)\rho_0$, a good incumbent is re-elected with probability $\rho_g = t + (1-t)\rho_0$. A bad incumbent's effort does not depend on ρ_0, which does not interact with e_1, so $\mathbb{E}_c[e_1^*] = t - t^2/4$ as in (14), and $\mathbb{E}_c[\rho_b] = \frac{t^2}{2}\big(1 - \ln\frac{t}{2}\big) + (1-t)\rho_0$. Both formulas hold on $[0,1]$, where the full-effort region $c \le t/2$ lies inside the support of c. Their derivatives are

$$\rho_g' = 1 - \rho_0, \qquad \frac{d}{dt}\mathbb{E}_c[e_1^*] = 1 - \frac{t}{2}, \qquad \frac{d}{dt}\mathbb{E}_c[\rho_b] = t\Big(\frac{1}{2} - \ln\frac{t}{2}\Big) - \rho_0, \tag{82}$$

the last because $\frac{d}{dt}\big[\frac{t^2}{2}(1 - \ln\frac{t}{2})\big] = t\big(1 - \ln\frac{t}{2}\big) - \frac{t}{2}$.

Step 2: the slope of g. Differentiating (5) term by term,

$$g'(t) = \gamma(1-\rho_0)\,(\tau_2 - \gamma\tau_1) \;+\; (1-\gamma)\,\tau_1\big[k(t) + \gamma\rho_0\big], \qquad k(t) \equiv 1 - \frac{t}{2} - \gamma\,t\Big(\frac{1}{2} - \ln\frac{t}{2}\Big). \tag{83}$$

The first term is the value of re-electing good incumbents more often: each additional re-election of a good type yields τ_2 instead of a challenger's expected $\gamma\tau_1$. The second collects the two effects of scrutiny on bad incumbents: more effort, $1 - t/2$, and more of them re-elected, each forgoing a challenger worth $\gamma\tau_1$ in expectation, which is the negative term in γ. Since $t\ln t \to 0$ as $t \to 0$, $k(0) = 1$ and g' is continuous on $[0,1]$.

Step 3. Only k depends on t, and

$$k'(t) = -\frac{1}{2} + \gamma\Big(\frac{1}{2} + \ln\frac{t}{2}\Big). \tag{84}$$

For $t \le 1$, $\ln\frac{t}{2} \le -\ln 2$, so $\frac{1}{2} + \ln\frac{t}{2} \le \frac{1}{2} - \ln 2 < 0$ and $k'(t) < -\frac{1}{2}$. Hence $g'' = (1-\gamma)\tau_1 k' < 0$ on $(0,1]$: g is strictly concave, and g' is strictly decreasing on $[0,1]$.

Step 4. Since g' is strictly decreasing, its smallest value on $[0,1]$ is at $t = 1$, where $k(1) = \frac{1}{2} - \gamma(\frac{1}{2} + \ln 2)$. So g is strictly increasing on $[0,1]$ if and only if

$$g'(1) = \gamma(1-\rho_0)(\tau_2 - \gamma\tau_1) + (1-\gamma)\,\tau_1\Big[\frac{1}{2} - \gamma\Big(\frac{1}{2} + \ln 2\Big) + \gamma\rho_0\Big] > 0. \tag{85}$$

If $\rho_0 < 1$, the coefficient of τ_2 in (85), $\gamma(1-\rho_0)$, is positive, and solving (85) for τ_2 gives $\tau_2 > \underline{\tau}\,\tau_1$, with $\underline{\tau}$ as in the statement. To see that $\underline{\tau} < 1$, set $\tau_2 = \tau_1$ in (85). The terms in ρ_0 cancel, and

$$g'(1)\big|_{\tau_2=\tau_1} = (1-\gamma)\,\tau_1\Big[\frac{1}{2} - \gamma\Big(\ln 2 - \frac{1}{2}\Big)\Big] > 0, \tag{86}$$

since $\ln 2 - \frac{1}{2} < \frac{1}{2}$ and $\gamma < 1$. So $\tau_2 = \tau_1$ already satisfies the condition, and because $g'(1)$ is increasing in τ_2, so does every $\tau_2 \ge \tau_1$; hence $\underline{\tau} < 1$. If $\rho_0 = 1$, the first term of (85) vanishes and $g'(1) = (1-\gamma)\tau_1\big[\frac{1}{2} - \gamma(\ln 2 - \frac{1}{2})\big] > 0$, so g is strictly increasing whatever τ_2. □

Proof of part (i) of Proposition 7 It is convenient to read κ as a price. Write

$$\mathcal{D}(t) \;=\; t\ln\frac{t}{\beta} + (1-t)\ln\frac{1-t}{1-\beta} \;\ge\; 0 \tag{87}$$

for the steering effort a voter exerts in moving his diet from the default β to t — zero when $t = \beta$, and larger the further he pulls away. The voter's problem is then

$$\Pi(t,\kappa) = \underbrace{t\ln\alpha + (1-t)\ln(1-\alpha)}_{\text{content payoff}} - \underbrace{\kappa\,\mathcal{D}(t)}_{\text{steering bill}}, \qquad v(\alpha;\kappa,\beta) = \max_{t\in[0,1]}\Pi(t,\kappa), \tag{88}$$

so κ is the price per unit of steering effort and $\mathcal{D}(t)$ is the quantity purchased. Unbundling is a fall in that price, and the private gain is the corresponding gain in consumer surplus.

Step 1. Π is strictly concave in t, since $\partial^2\Pi/\partial t^2 = -\kappa\left[\frac{1}{t} + \frac{1}{1-t}\right] < 0$, and $\partial\Pi/\partial t$ diverges to $+\infty$ as $t \to 0$ and to $-\infty$ as $t \to 1$. The optimal diet $t^*(\kappa)$ is therefore interior and unique, characterized by (21).

Step 2. Raising κ affects v in two ways: it makes the steering already being done more expensive, and it induces the voter to steer less. The second effect is weaker, because the voter was optimizing. Formally, Π is affine in κ with coefficient $-\mathcal{D}(t)$, so v is a maximum of affine functions of κ — hence convex — and, the maximizer being unique, differentiable, with

$$\frac{\partial v}{\partial \kappa} \;=\; -\,\mathcal{D}\big(t^*(\kappa)\big). \tag{89}$$

The voter loses, per unit increase in the price, exactly the amount of steering he is already doing.

Step 3. $\mathcal{D}(t) \geq 0$ with equality only at $t = \beta$, so the price matters to a voter if and only if he departs from the default. By (21),

$$\ln \frac{t_n^*}{1 - t_n^*} = \frac{1}{\kappa} \ln \frac{\alpha}{1-\alpha} + \ln \frac{\beta}{1-\beta}, \tag{90}$$

so $t_n^* = \beta$ exactly when $\ln \frac{\alpha}{1-\alpha} = 0$, that is $\alpha = \frac{1}{2}$ — whatever the price and whatever the default. A voter choosing to divide his time equally between news and entertainment wants the default bundle and does not buy any steering. Every other voter buys some, and pays for it. Hence $\partial v/\partial\kappa < 0$ for all $\alpha \neq \frac{1}{2}$, and $\partial v/\partial\kappa = 0$ at $\alpha = \frac{1}{2}$.

Step 4. Integrating Step 2 between the two prices,

$$\underbrace{v(\alpha;\kappa_U,\beta) - v(\alpha;\kappa_B,\beta)}_{\text{private gain}} \;=\; \int_{\kappa_U}^{\kappa_B} \mathcal{D}\big(t^*(\kappa)\big)\, d\kappa \;>\; 0 \qquad \text{for } \alpha \neq \tfrac{1}{2}, \tag{91}$$

and the gain is zero at $\alpha = \frac{1}{2}$. The right-hand side of (91) is the area under the voter's demand curve for steering between the old and new prices. The voter steers less the higher the price of steering: by the log-odds equation above, raising κ shrinks the term $\frac{1}{\kappa}\ln\frac{\alpha}{1-\alpha}$, which moves t_n^* back towards β, and $\mathcal{D}$ falls as the diet approaches β. The private gain from unbundling is therefore the consumer-surplus gain from a fall in the price of steering. It is larger the further the voter's taste lies from $\frac{1}{2}$, on either side, because such a voter steers more at every price. No restriction on β was used: the gain is positive under any default mix.

Step 5. Let $\beta = \frac{1}{2}$. Then by (29), $v = \kappa \ln\left[\frac{1}{2}\alpha^{1/\kappa} + \frac{1}{2}(1-\alpha)^{1/\kappa}\right]$ is a symmetric function of the pair $(\alpha, 1-\alpha)$: replacing α by $1-\alpha$ merely interchanges its two arguments. Hence $v(1-\alpha;\kappa) = v(\alpha;\kappa)$ for every κ, and the gain inherits the symmetry. This is the only step that uses $\beta = \frac{1}{2}$. In the asymmetric default, the two arguments involve unequal weights, and, as a result, the symmetry fails.

Step 6. By Steps 3–4, the gain is non-negative everywhere and strictly positive except at the single point $\alpha = \frac{1}{2}$. Hence $G(F) = \mathbb{E}_F\big[v(\alpha;\kappa_U) - v(\alpha;\kappa_B)\big] > 0$ for every taste distribution other than the point mass at $\frac{1}{2}$.

Proof of part (ii) of Proposition 7. Suppose $\Delta(F) \geq 0$, for any default β. By Lemma 12, g is increasing, so $g(\tilde{t}_n^B + \Delta) - g(\tilde{t}_n^B) \geq 0$. Adding $G(F) > 0$ from part (i) gives $\Delta W > 0$ whenever F is not the point mass at $\frac{1}{2}$. The two channels then reinforce

rather than conflict with each other, and there is no threshold. At the point mass, $\Delta = 0$ and $G = 0$, so $\Delta W = 0$.

Suppose $\beta = \frac{1}{2}$. Then if F is symmetric around $\frac{1}{2}$, Proposition 6(ii) gives $\Delta(F) = 0$; the public term in (31) becomes zero, and $\Delta W = G(F)$, which is strictly positive unless F is the point mass at $\frac{1}{2}$. The Beta density is symmetric precisely when $\mu = \frac{1}{2}$, for every ϑ. Furthermore, in the Beta family the threshold of Proposition 6(iii) is $\mu^* = \frac{1}{2}$, so $\Delta(F) \geq 0$ for every $\mu \geq \frac{1}{2}$.

Part (iii) differs from part (ii) because the two channels aggregate differently. The private gain is positive for every voter, so it stays positive when averaged over any electorate. The change in scrutiny has opposite signs on the two sides of $\frac{1}{2}$ (Lemma 4(ii)), so it can cancel out, and with an evenly mixed default it cancels exactly in a balanced electorate. This is why a balanced electorate is unambiguously better off, however large or small each channel is.

Proof of part (iii) of Proposition 7. Rearranging (31), $\Delta W < 0$ if and only if

$$g\big(\tilde{t}_n^B\big) - g\big(\tilde{t}_n^B + \Delta(F)\big) > G(F), \tag{92}$$

which is the first statement. Differentiability is not required. By part (i) the right-hand side is strictly positive, so (92) can hold only if the loss of accountability on the left is strictly positive too — as it is, since $\Delta(F) < 0$ and g is increasing.

For the second statement, note first that $S(F)$ is well defined and finite: $G(F) > 0$ by part (i), and $|\Delta(F)| > 0$ by hypothesis. By Lemma 12, g is continuously differentiable, so, since $\Delta(F) < 0$, the mean value theorem provides $\xi \in \big(\tilde{t}_n^B + \Delta(F),\, \tilde{t}_n^B\big)$ with

$$g\big(\tilde{t}_n^B\big) - g\big(\tilde{t}_n^B + \Delta(F)\big) = g'(\xi)\,\big|\Delta(F)\big|. \tag{93}$$

This step uses only the differentiability of g, not its functional form. Dividing by $|\Delta(F)| > 0$ turns (92) into $g'(\xi) > S(F)$. Both sides are measured in content utility per unit of aggregate news share, by the normalization of Section 3.2. Finally, because g is concave (Lemma 12), g' is decreasing, and $\tilde{t}_n^B + \Delta(F) < \xi < \tilde{t}_n^B$ gives

$$g'\big(\tilde{t}_n^B\big) \;\leq\; g'(\xi) \;\leq\; g'\big(\tilde{t}_n^B + \Delta(F)\big), \tag{94}$$

so $g'(\tilde{t}_n^B) > S(F)$ is sufficient for welfare to fall, and $g'(\tilde{t}_n^B + \Delta(F)) > S(F)$ is necessary. □